344 – Section 8
Wind turbine design

Choices

25-MW turbine:

- 3-bladed
- 1-bladed
- On Mars
- Downwind
- (Stall regulated)

- Floating
- AI-designed

- Other idea?

Small scale:

- 3D printed scale
- DOE Collegiate Wind Competition

Advanced:

- Vertical axis
- Kite/Airborne

Some steps for preliminary design

25 MW design – Preliminary design

Setup:

- Start a report, Git pull the data-ref to get sample_code/aerodesign.py
- Copy the 22 MW folder to a 25 MW folder, Make the structure stiff (no DOF in ElastoDyn input file)
- Make sure you can run OpenFAST on the new input files.

Environment:

- Chose an approximate location, and figure out the environmental conditions there

Aerodynamics:

- Decide on a number of blade (for fun),
- Decide whether you'll have high specific power or not, then find rough CP values (CP opt and rated)
- Figure out the approximate hub height and rotor radius using rule of thumb
- Find potential design constraints (max chord, and max tip speed, max tip height, max weight?)
- Roughly estimate the TSR
- Choose an airfoil, perform a preliminary blade design based on HW3
- Run this blade design in OpenFAST at your designed TSR. Calculate your CP

25 MW design – Preliminary design

Environment:

- Generate a turbulence box suitable for your design
- Generate a wind step input file
- Perform an environmental impact assessment (endangered species, birds, landscape, stakeholders, community)

Aerodynamics (continued):

- Run your blade design in OpenFAST at your designed TSR. Calculate your CP.
- Make sure the blade satisfy the constraints (Max tip-speed, max chord)
- Iterate on your design if you are not happy with your CP, or to account for the constraints.

Controls:

- Calculate your Controller parameters and your region 2 proportionality constant
- Activate ServoDyn
- Run an OpenFAST simulation with varying wind (and ServoDyn on), verify that the generator works properly.
    - Wind speeds

25 MW design – Preliminary design

Structure:

- Estimate your maximum thrust force
- Scale the blade stiffness and mass
- Scale the tower stiffness and mass
- Scale some inertias
- See if the model runs…
- Estimate natural frequencies (with your scripts or with linearization, or ACDC). Make sure the tower frequency is not around 1P or 3P
- Estimate the loading needed to have a tower strike.
- Compute the maximum tower top deflection, bending stresses for maximum thrust at the blade root and tower root
- Perform a couple of turbulent simulation, and compute the damage equivalent loads for the blade root bending moment (m=10) and tower bottom bending moment (m=4)

Some studies ideas

- Estimate the energy output at your site (Annual Energy Production).
- Perform an environmental impact assessment (endangered species, birds, landscape, stakeholders, community)
- Continue some of the homework
- Develop more structural models
- Do some simple structural analyses (maximum tower and blade deflections)
- Design a controller using ROSCO or do it yourself with a simple model

- Go with your interests, see what simple analyses / engineering questions you can answer.

Day 1

25 MW design – Preliminary design

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Preliminary site assessment

Environmental conditions

Environmental conditions

Essential:

- Air density

- Mean wind speed as function of height or rough idea of wind condition

Potentially:

- Probability density function as function of wind speed

- Wind rose

Preliminary aerodynamic design

Rule of thumbs

We write P the power produced by a wind turbine. We can define the power coefficient as the ratio between the power generated and the power available in the wind: **D**

$$C_P \triangleq \frac{P}{P_{\text{wind}}} = \frac{P}{\frac{1}{2}\rho A U_0^3} \qquad (< \frac{16}{27}) \tag{1.2}$$

In practice, we like to write this formula as:

$$P = \frac{1}{2}\rho A U_0^3 C_P \tag{1.3}$$

$$P, \text{rated} \approx \{1., 1.2\} R^2$$

$$C_{P,\text{rated}} \approx \{0.25, 0.3\}$$

$$R = \sqrt{\frac{P_{\text{rated}}}{\frac{1}{2}\rho V_{\text{rated}}^3 \pi C_{P,\text{rated}}}}$$

Specific power

To compare different wind turbine design, we also use the notion of specific power, which corresponds to the power per area:

$$\text{Specific power} \triangleq \frac{P}{A} = \frac{1}{2}\rho U_0^3 C_P \tag{1.13}$$

You can look at turbine specific power in the US here. Specific power range around [200 – 400] W/m^2.

https://emp.lbl.gov/specific-power

Question 1.3: Do we want to use a high or low specific power?

▸ A high specific power is linked to a high C_P, which means it's a highly efficient wind turbine design. Such designs are desired to extra as much energy from the wind as possible. In regions with a lot of wind farm (high wind energy penetration), electricity prices can become close to zero or even negative at high wind because a lot of wind farms are operating at the same time. There is therefore a drive to generate power at lower wind speed (shifting the power curve to the left). To achieve this, we design wind turbines at a low specific power, lower C_P and larger rotor area. The choice is therefore a trade off depending on the market in which the turbine operates.

Maximum performance possible

We need to maximize the power in each annulus.

From BET we have:

C_P in annulus:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = \frac{1}{2}\rho U_{\mathrm{rel}}^2 \Omega r B c \, c_t$$

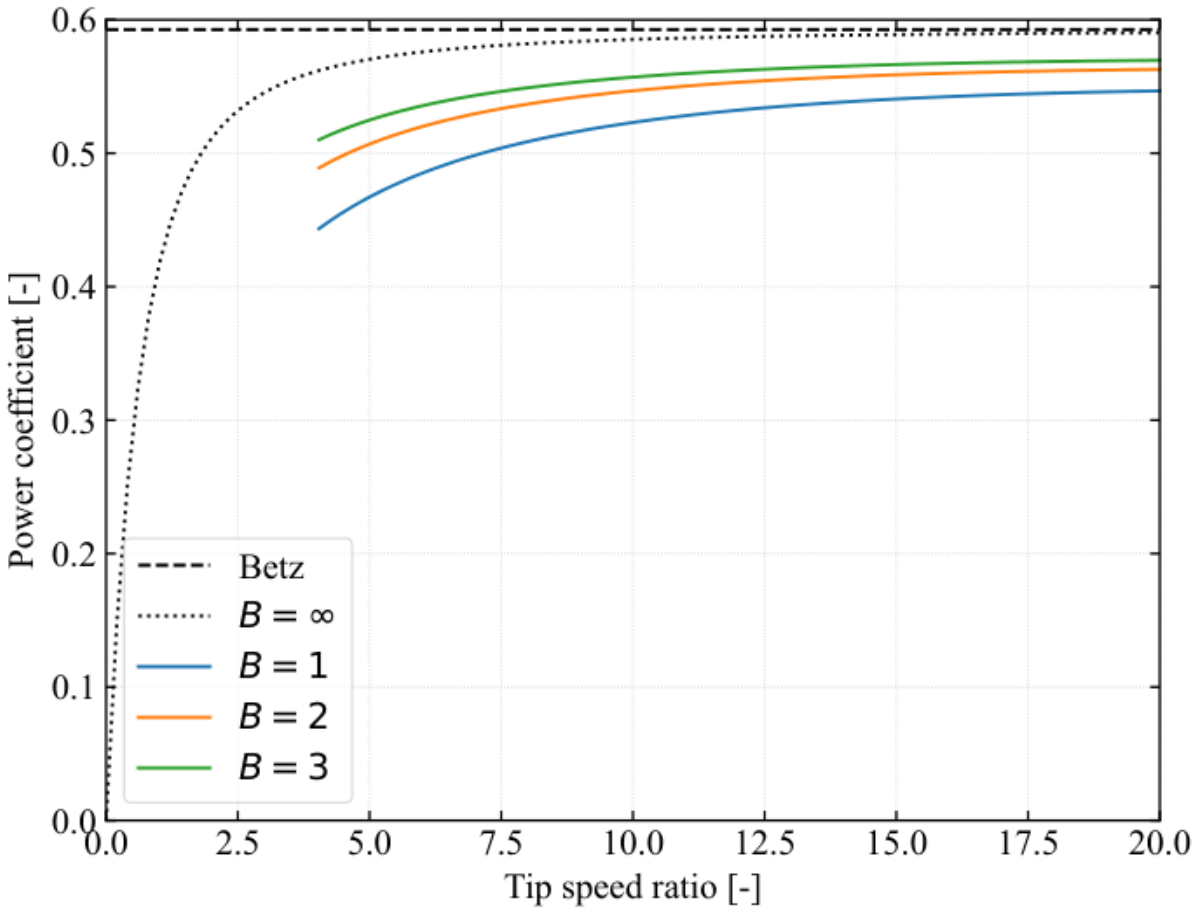
$$\frac{\mathrm{d}C_P}{\mathrm{d}r} = \frac{U_{\mathrm{rel}}^2}{U_0^2}\frac{\Omega r}{U_0}\frac{Bc}{\pi R^2} c_t$$

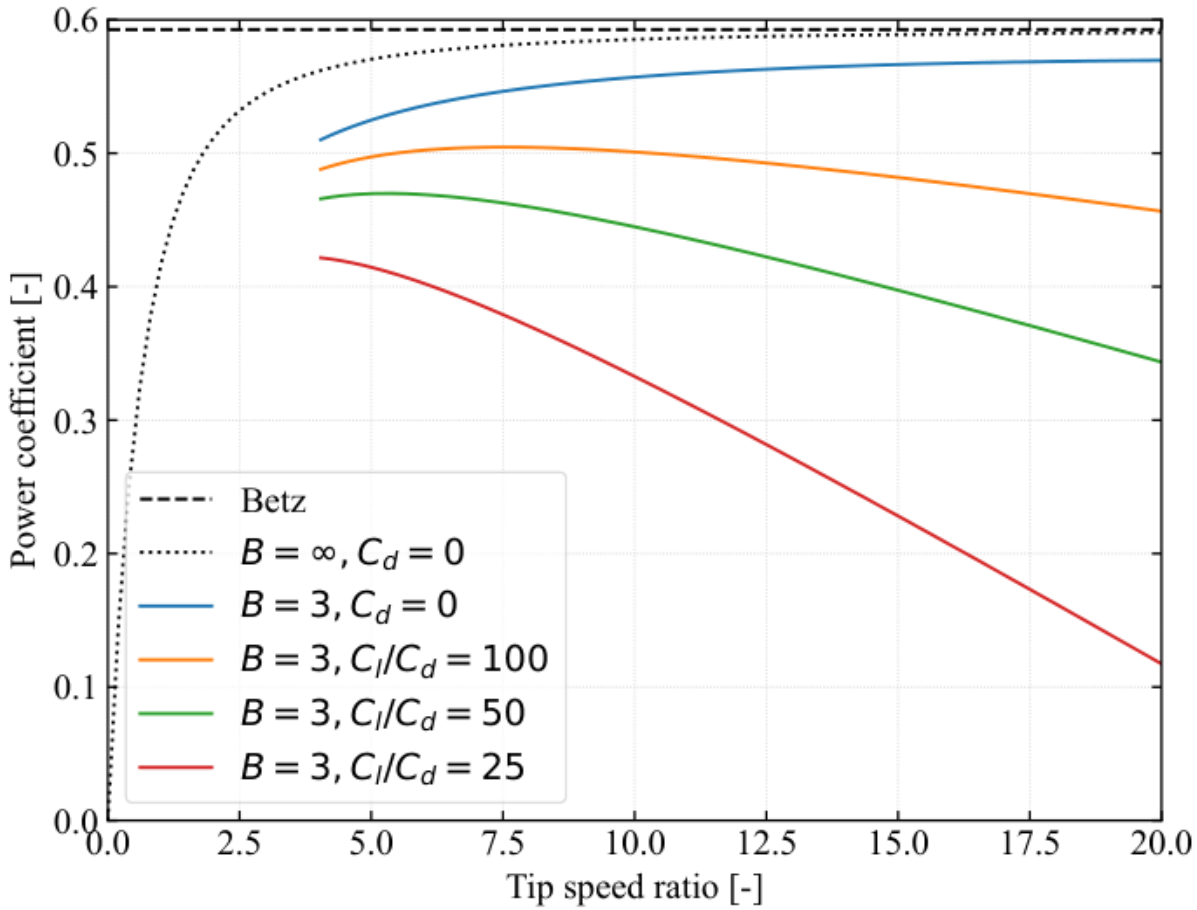
$$\frac{\mathrm{d}C_P}{\mathrm{d}\lambda_r} = \frac{2}{\lambda^2}\frac{U_{\mathrm{rel}}^2}{U_0^2}\lambda_r^2 \sigma c_t \approx f(\lambda, \lambda_r, \sigma c_t)$$

We can combine with the Momentum theory equations and try to find the maximum Cp for each annulus (written dCp/dlambda_r above).

Maximum power as function of TSR, B, Cl/Cd

Use provided function in aerodesign.py :
CP_lambda_B_ClCd_Wilson





Solidity

Note, from BEM, we have:

$$a = \frac{1}{4F\sin^2\phi/(\sigma c_n) + 1}$$

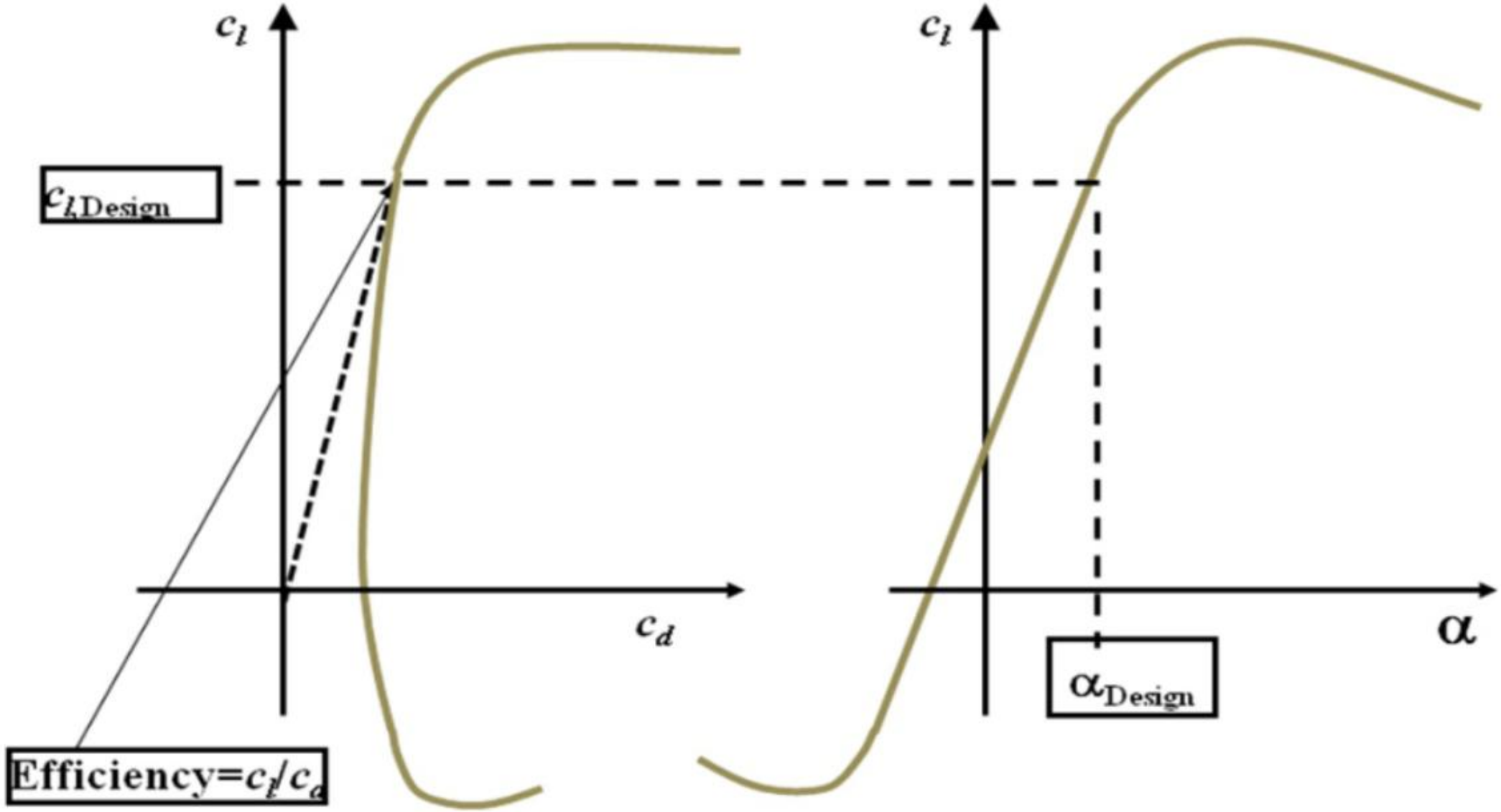
Therefore if $a = 1/3$, we get:

$$\sigma = \frac{2F\sin^2\phi}{c_n}$$

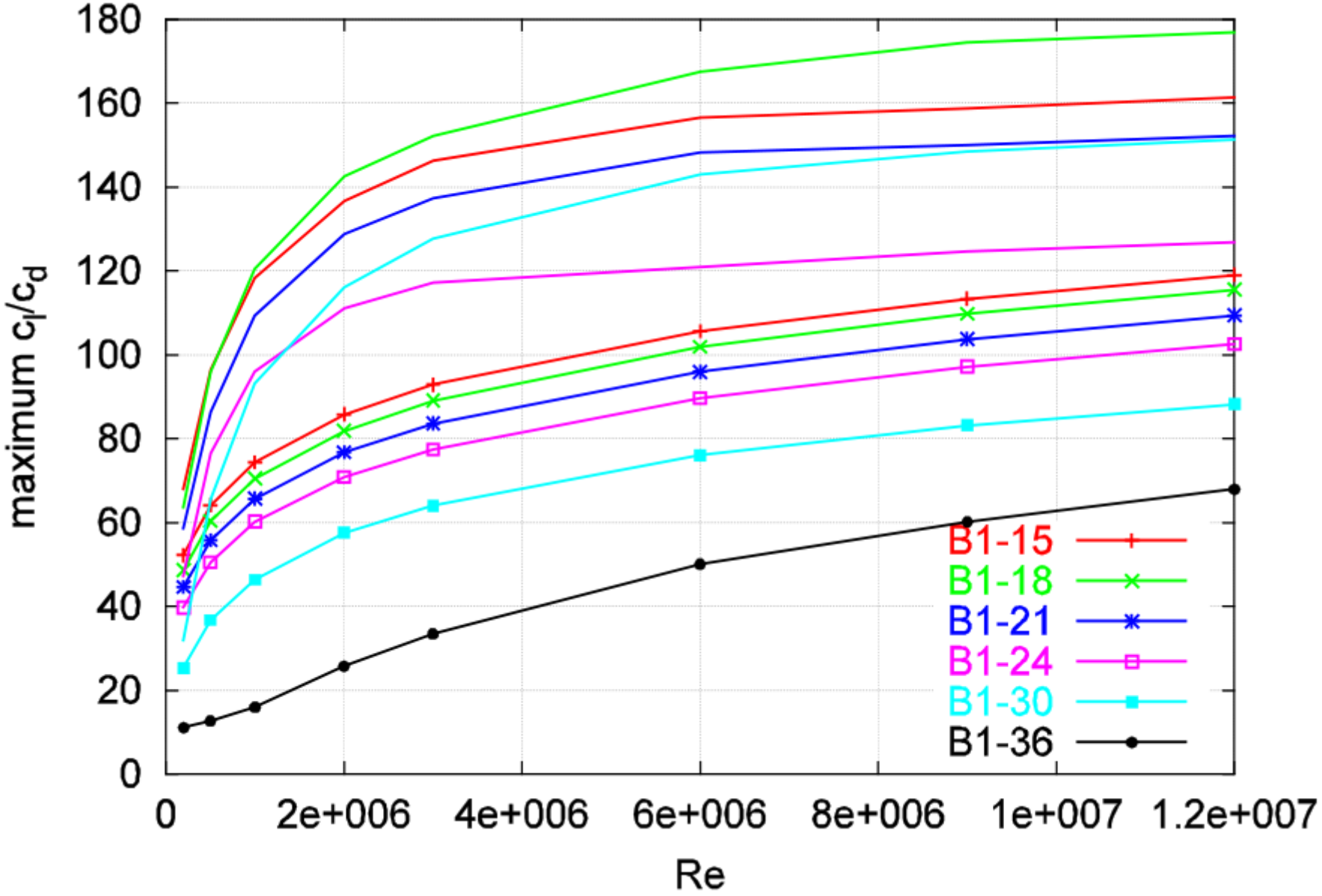
Thus:

- Higher C_l leads to smaller solidity (i.e. smaller chord)
- Higher λ leads to smaller solidity

Choice of airfoil

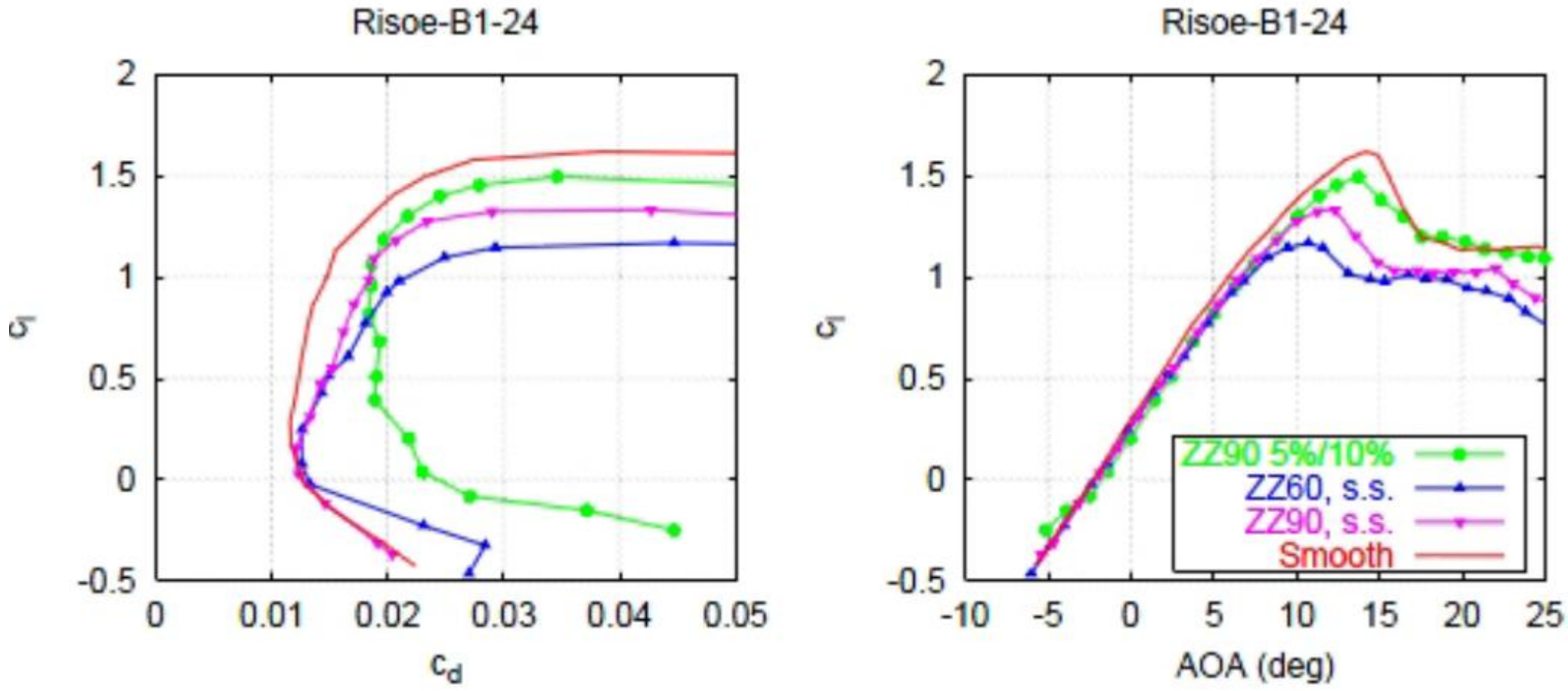


Airfoil performance – As function of Re

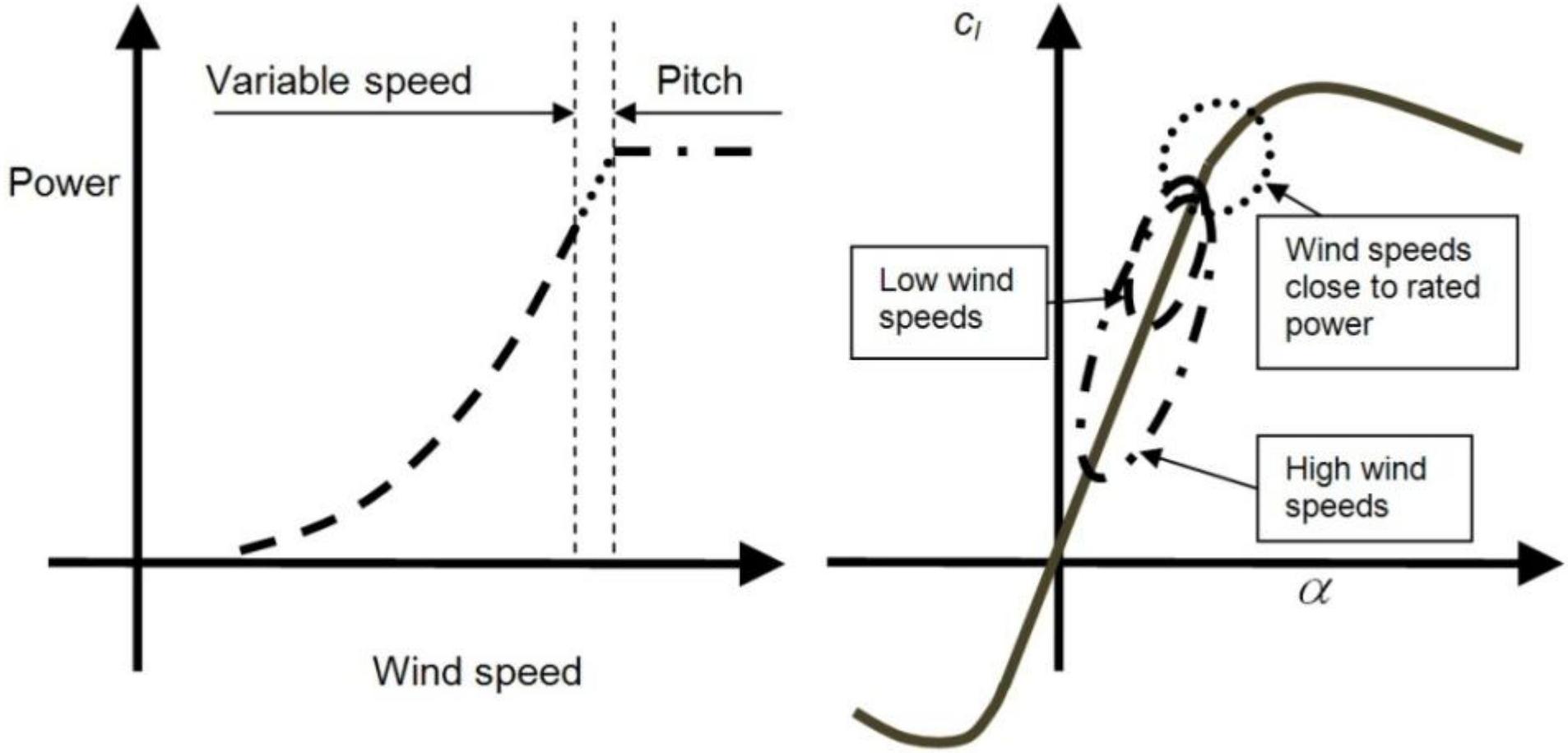


Airfoil performance - Soiling

Airfoil performance - Soiling

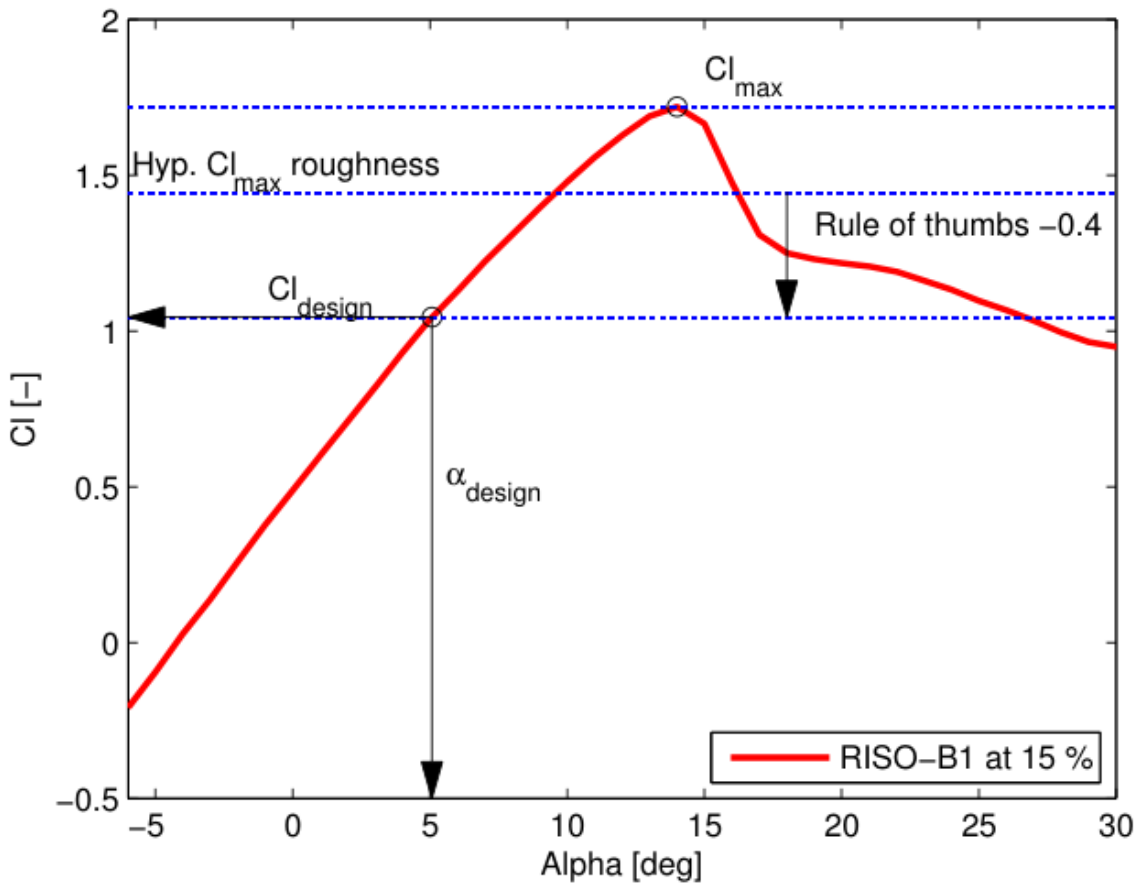


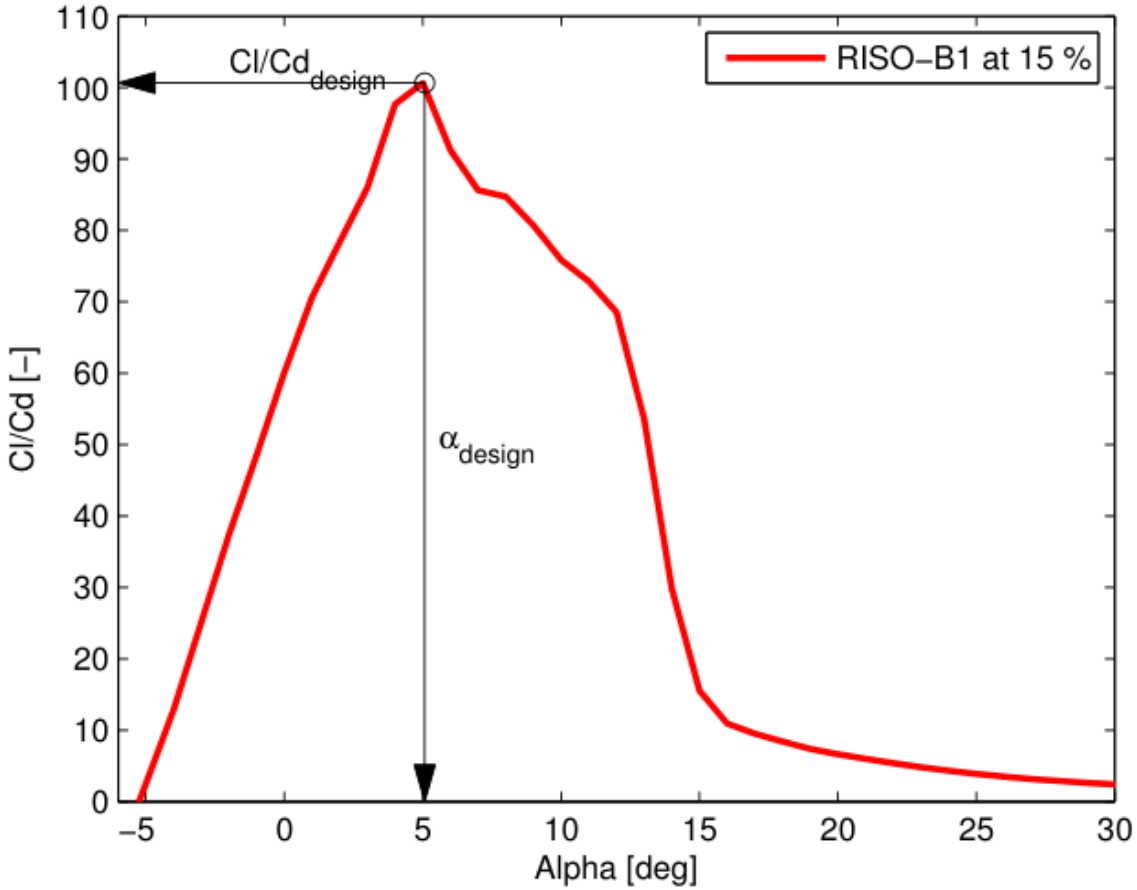
Airfoil and Control Strategy



Finding design point – rule of thumb

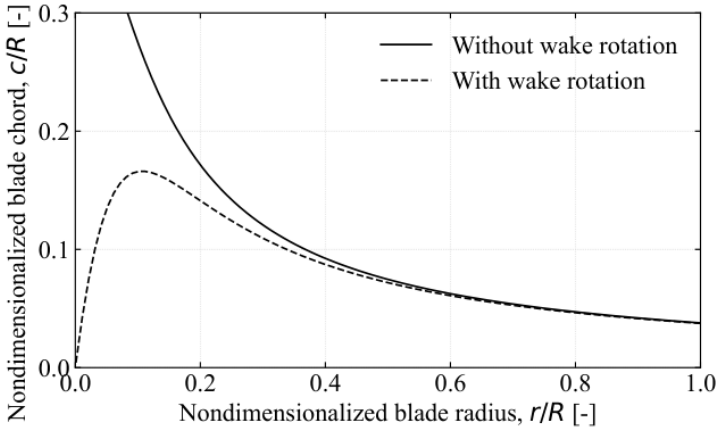
Rule of Thumb: Cl design = Clmax – 0.4

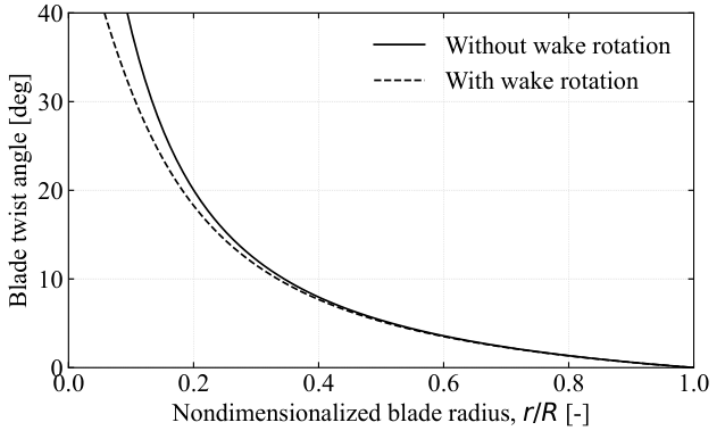




Chord and twist

Use provided function: planform_ClCd



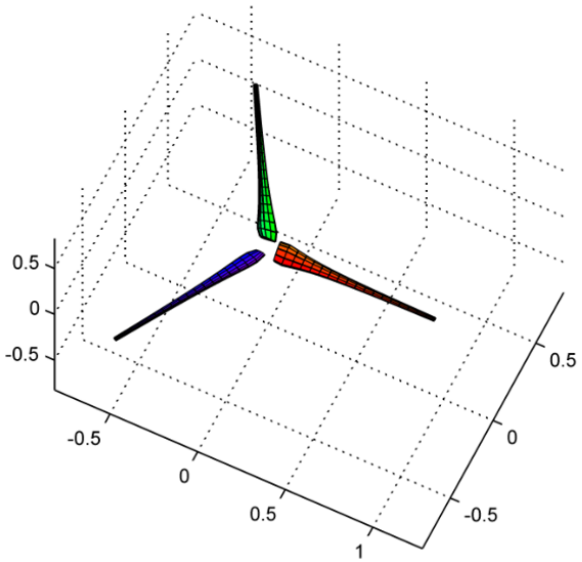


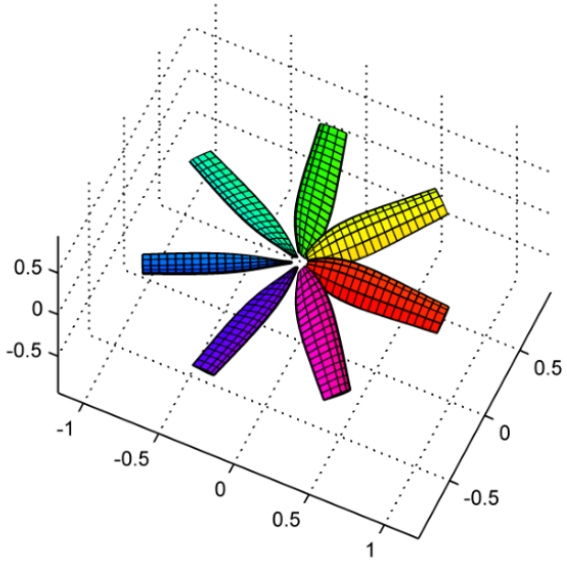
$$\phi(r) = \tan^{-1} \frac{1-a}{(1+a')\lambda_r}$$

$$c_n(r) = C_{l,\text{design}} \cos\phi + C_{d,\text{design}} \sin\phi$$

$$F(r) = \frac{2}{\pi} \cos^{-1}\left(\exp\left[-\frac{B}{2}\frac{(R-r)}{r\sin\phi}\right]\right)$$

$$c(r) = \frac{a}{1-a}\frac{8\pi r F \sin^2\phi}{B c_n}$$





Day 2

25 MW design – Preliminary design

Environment:

- Generate a turbulence box suitable for your design
- Generate a wind step input file

Aerodynamics (continued):

- Run your blade design in OpenFAST at your designed TSR. Calculate your CP.
- Make sure the blade satisfy the constraints (Max tip-speed, max chord)
- Iterate on your design if you are not happy with your CP, or to account for the constraints.

Controls:

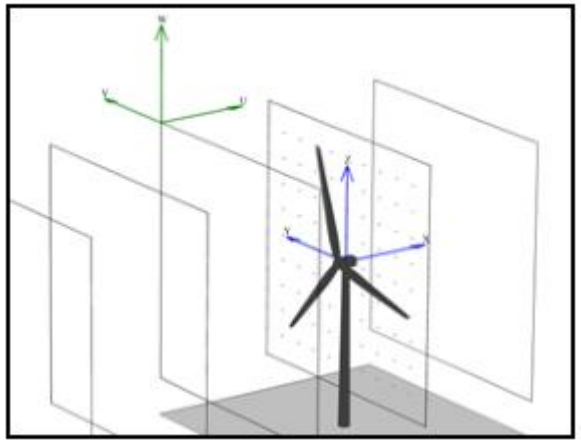
- Calculate your Controller parameters and your region 2 proportionality constant
- Activate ServoDyn
- Run an OpenFAST simulation with varying wind (and ServoDyn on), verify that the generator works properly.
    - Wind speeds
    - Turbulence

TurbSim

TurbSim

- NREL code
- Creates **multiple time series** of **turbulent** wind across rotor plane that are **correlated** according to the **distance** and that are consistent with a chosen **power spectral density** (Kaimal or von Karman)
- Two ways to look at it:
  - Time series at the rotor plane (spacing: Δt)
  - Multiple planes (spacing $\Delta x = U_0 \Delta t$)
- See TurbSim Manual in:

  02_Wind/WindSimulation



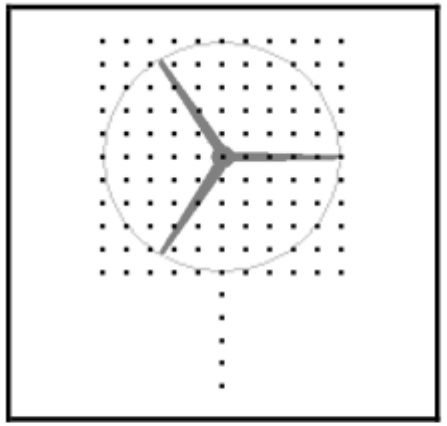


Figure 4. Example of tower points below a rectangular grid

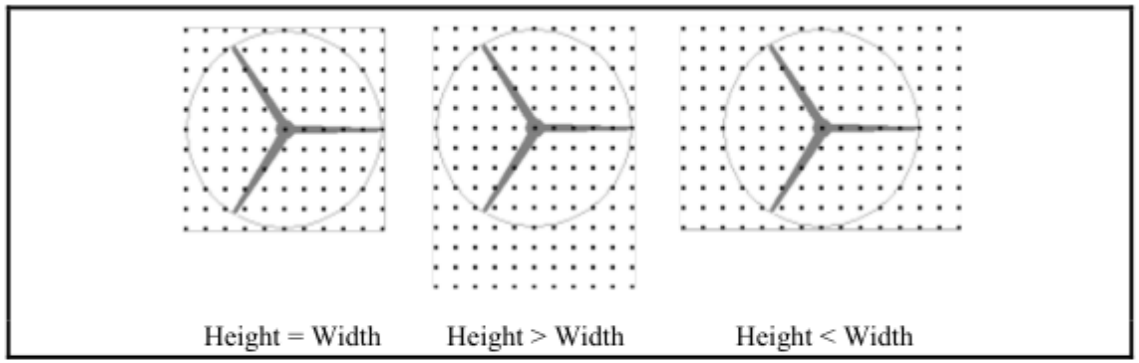
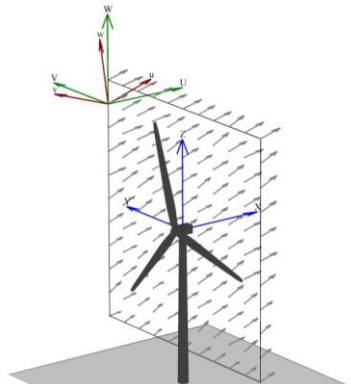


Figure 5. Example grid and rotor placements: the circles pictured here are the rotor diameters assumed by TurbSim; the actual rotor diameter(s) will be smaller than pictured

Important:

- Use BTS format and ask for tower points (lollipop)
- Use custom coherence (double check with pyDatView)
- The input "HubHeight" is the middle of the box, it doesn't have to be the hub height.

Coordinates of a TurbSim wind field with 15° horizontal and 8° vertical mean flow angles



Prescribed Uniform Wind – E.g. Wind Steps

InflowWind:

- WindType =2 (Uniform)
- FileName_Uni = “WindStepPowerCurve.wnd”

```python
from openfast_toolbox.case_generation.case_gen import createStepWind

createStepWind('WindStepPowerCurve3.wnd',WSstep=1.,WSmin=2,WSmax=15,tstep=100,dt=0.5)
```

```
!Wind file.
!Time  Wind     Wind    Vert.       Horiz.      Vert.       LinV        Gust
!      Speed    Dir     Speed       Shear       Shear       Shear       Speed
0.0 2.0 0.0 0.0 0.0 0.0 0.0 0.0
99.5 2.0 0.0 0.0 0.0 0.0 0.0 0.0
100.0 3.0 0.0 0.0 0.0 0.0 0.0 0.0
199.5 3.0 0.0 0.0 0.0 0.0 0.0 0.0
200.0 4.0 0.0 0.0 0.0 0.0 0.0 0.0
299.5 4.0 0.0 0.0 0.0 0.0 0.0 0.0
300.0 5.0 0.0 0.0 0.0 0.0 0.0 0.0
399.5 5.0 0.0 0.0 0.0 0.0 0.0 0.0
400.0 6.0 0.0 0.0 0.0 0.0 0.0 0.0
499.5 6.0 0.0 0.0 0.0 0.0 0.0 0.0
500.0 7.0 0.0 0.0 0.0 0.0 0.0 0.0
599.5 7.0 0.0 0.0 0.0 0.0 0.0 0.0
600.0 8.0 0.0 0.0 0.0 0.0 0.0 0.0
699.5 8.0 0.0 0.0 0.0 0.0 0.0 0.0
700.0 9.0 0.0 0.0 0.0 0.0 0.0 0.0
799.5 9.0 0.0 0.0 0.0 0.0 0.0 0.0
800.0 10.0 0.0 0.0 0.0 0.0 0.0 0.0
899.5 10.0 0.0 0.0 0.0 0.0 0.0 0.0
900.0 11.0 0.0 0.0 0.0 0.0 0.0 0.0
999.5 11.0 0.0 0.0 0.0 0.0 0.0 0.0
1000.0 12.0 0.0 0.0 0.0 0.0 0.0 0.0
1099.5 12.0 0.0 0.0 0.0 0.0 0.0 0.0
1100.0 13.0 0.0 0.0 0.0 0.0 0.0 0.0
1199.5 13.0 0.0 0.0 0.0 0.0 0.0 0.0
1200.0 14.0 0.0 0.0 0.0 0.0 0.0 0.0
1299.5 14.0 0.0 0.0 0.0 0.0 0.0 0.0
1300.0 15.0 0.0 0.0 0.0 0.0 0.0 0.0
1500.0 15.0 0.0 0.0 0.0 0.0 0.0 0.0
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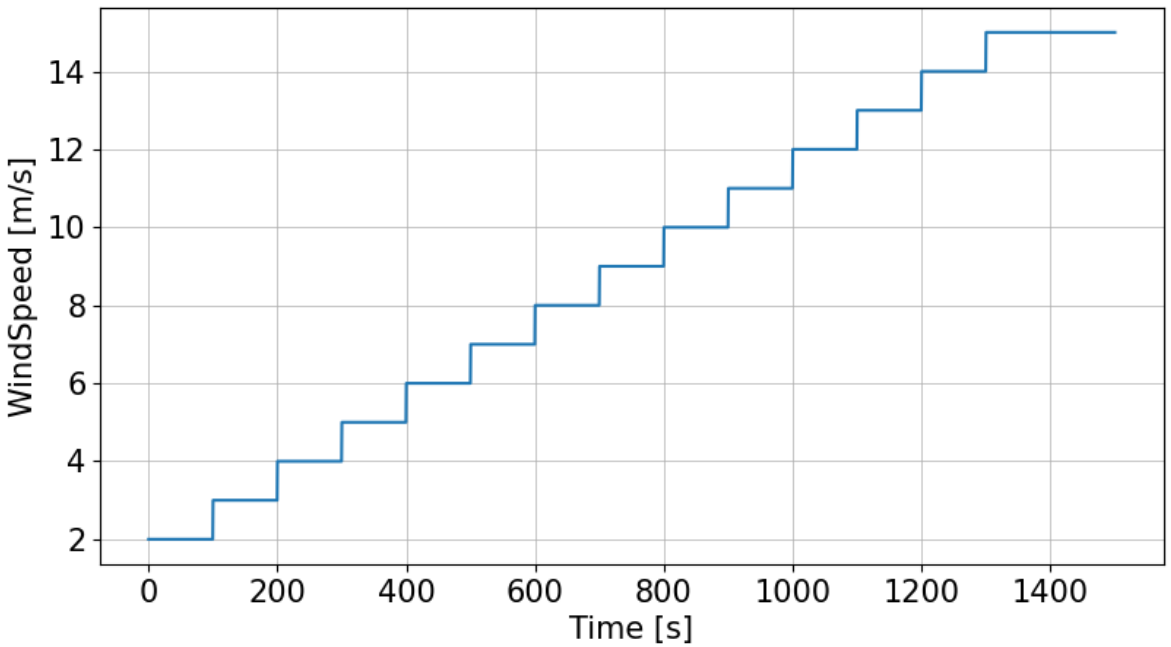
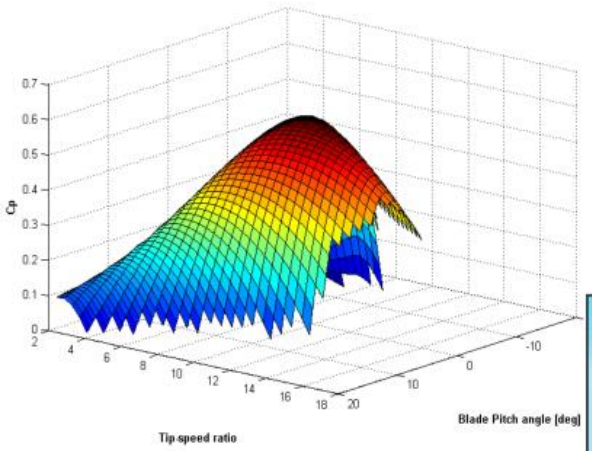


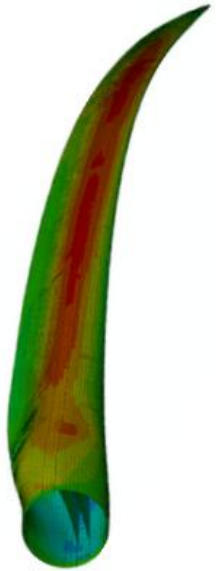
OpenFAST

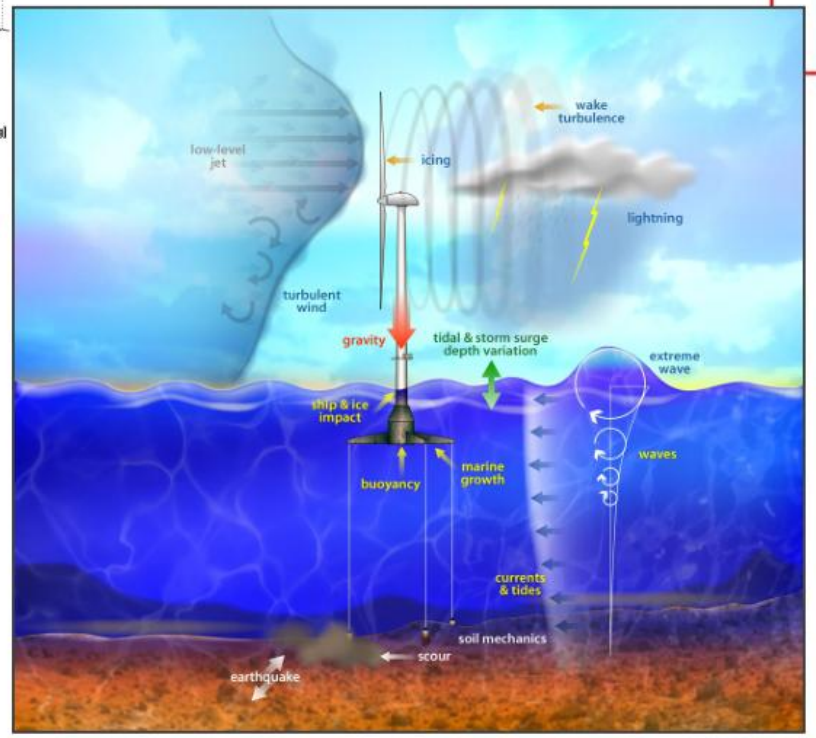
Physical Couplings

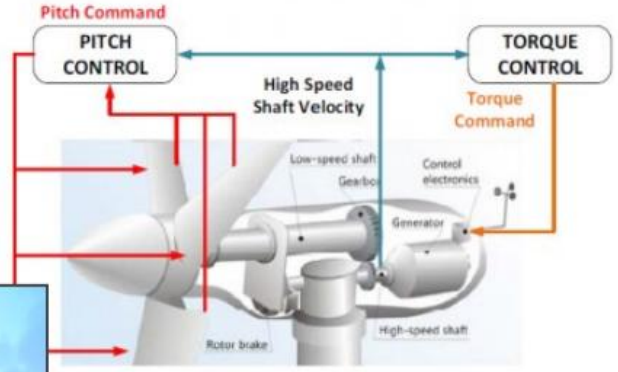


Aerodynamics

Structures



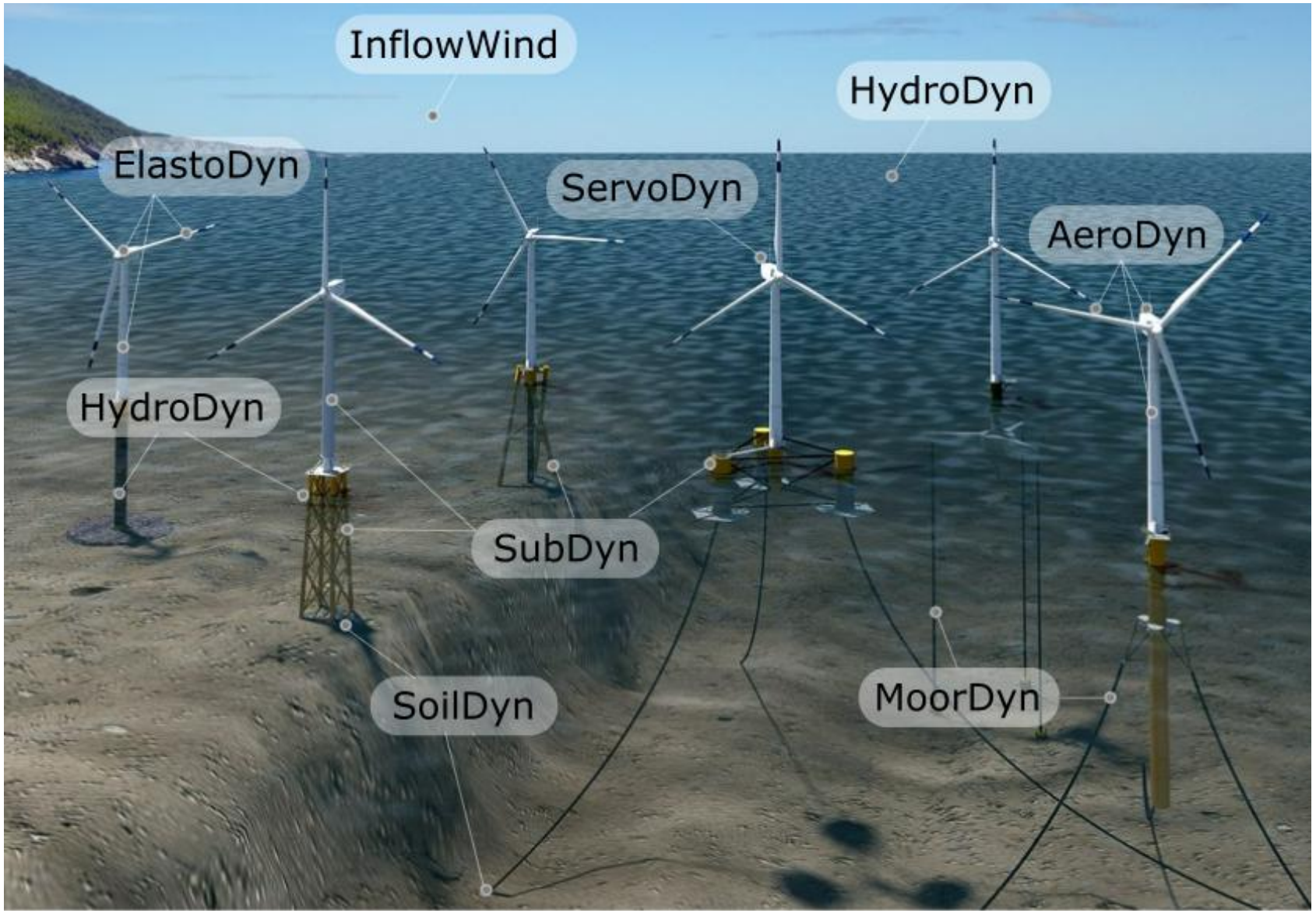




Controls

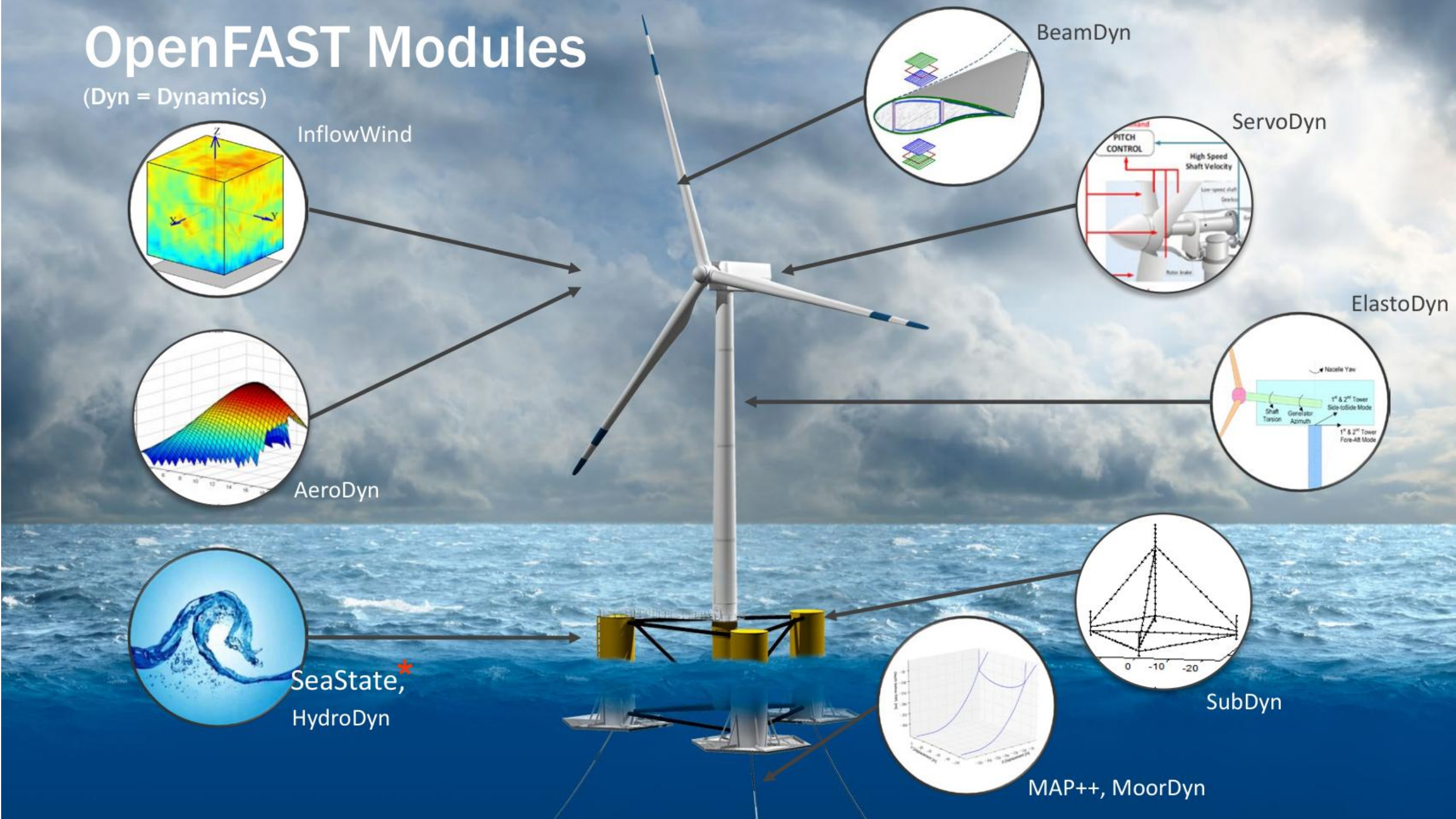
Hydrodynamics

OpenFAST Modules



OpenFAST Modules

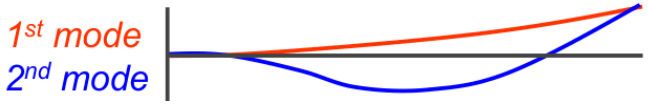
(Dyn = Dynamics)



Structural modules

ElastoDyn:

- Blades: shape function approach
- Tower: shape function approach

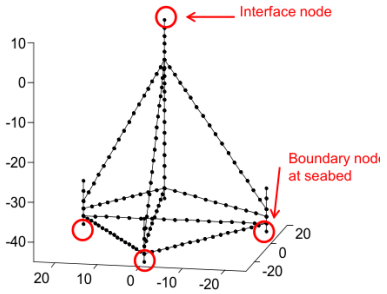


BeamDyn:

- Blades: geometrically exact beam theory (spectral finite element method)

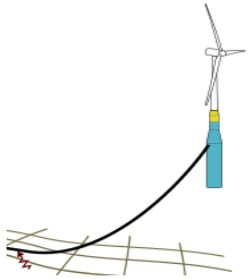
SubDyn:

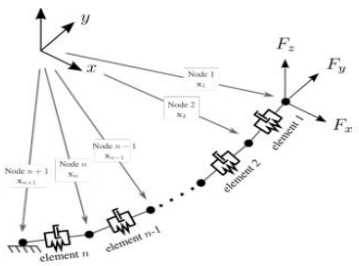
- Tower/Substructure: linear beam finite element method



MoorDyn

- Moorings: Lumped-Mass dynamics





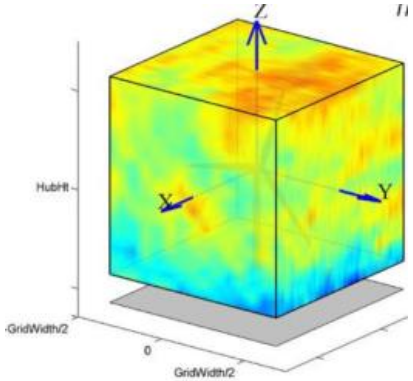
Environmental and External Force modules

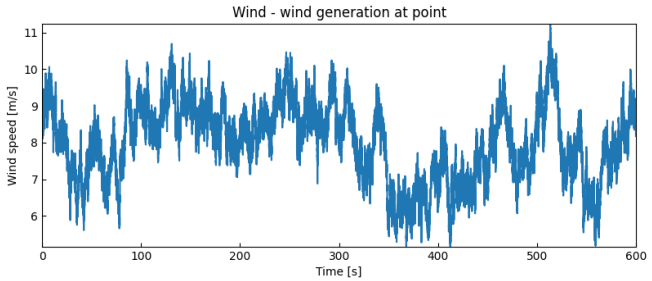
InflowWind:

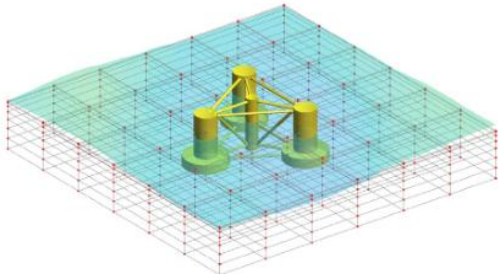
- Steady / turbulent wind

SeaState:

- Waves and current

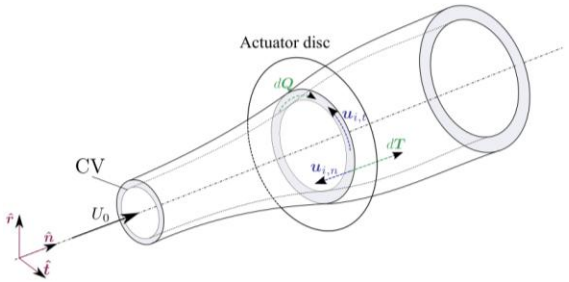


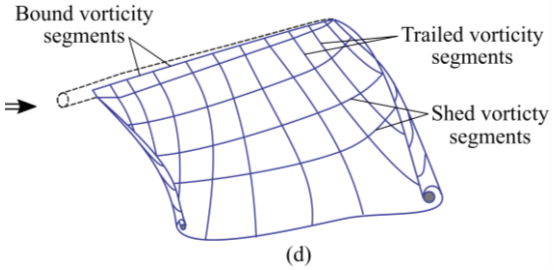




AeroDyn:

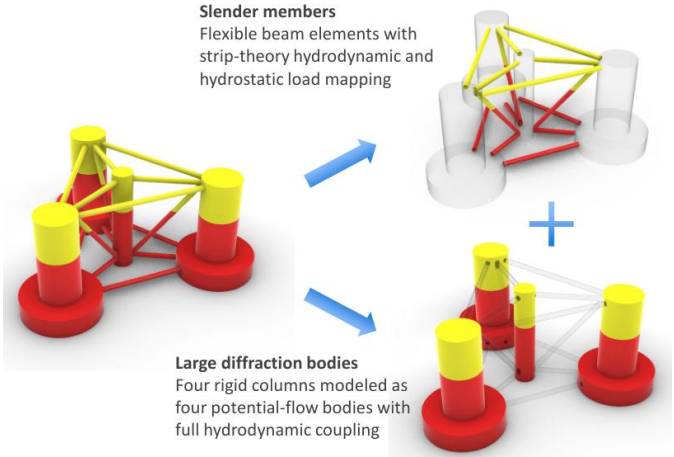
- Unsteady Blade Element Momentum
- Vortex method (OLAF)





HydroDyn:

- Morison force for slender structure
- Potential flow force for larger volume



Resources

Documentation:

- https://openfast.readthedocs.io/

Try the quick start: https://openfast.readthedocs.io/en/dev/source/working.html

NAWEA Workshop :

- https://drive.google.com/drive/folders/1zGNfGOOa0jLgSUNx9lfPtmx9EQt1TGfq
- https://nrel.github.io/WETOStack/workshops/nawea_2024.html#openfast-overview-practical-workshop

OpenFAST

Getting started

See OpenFAST workshop slides

OpenFAST

Tools

See OpenFAST workshop slides

Tools overview

Fundamentals (executables / libraries):

- **OpenFAST** realm: https://github.com/OpenFAST/openfast
  - Glue codes executables (use multiple modules): openFAST, FAST.farm, TurbSim
  - Driver executables (use a single module): aerodyn_driver, hydrodyn_driver
  - Libraries: Simulink, C++ library, DISCON (controller), more and more C-libraries wrapped with python

  They take input files and return outputs (files).
- **ROSCO**: opensource controller that can be used with OpenFAST
  https://github.com/NREL/ROSCO

Pre/post-processing and GUIs:

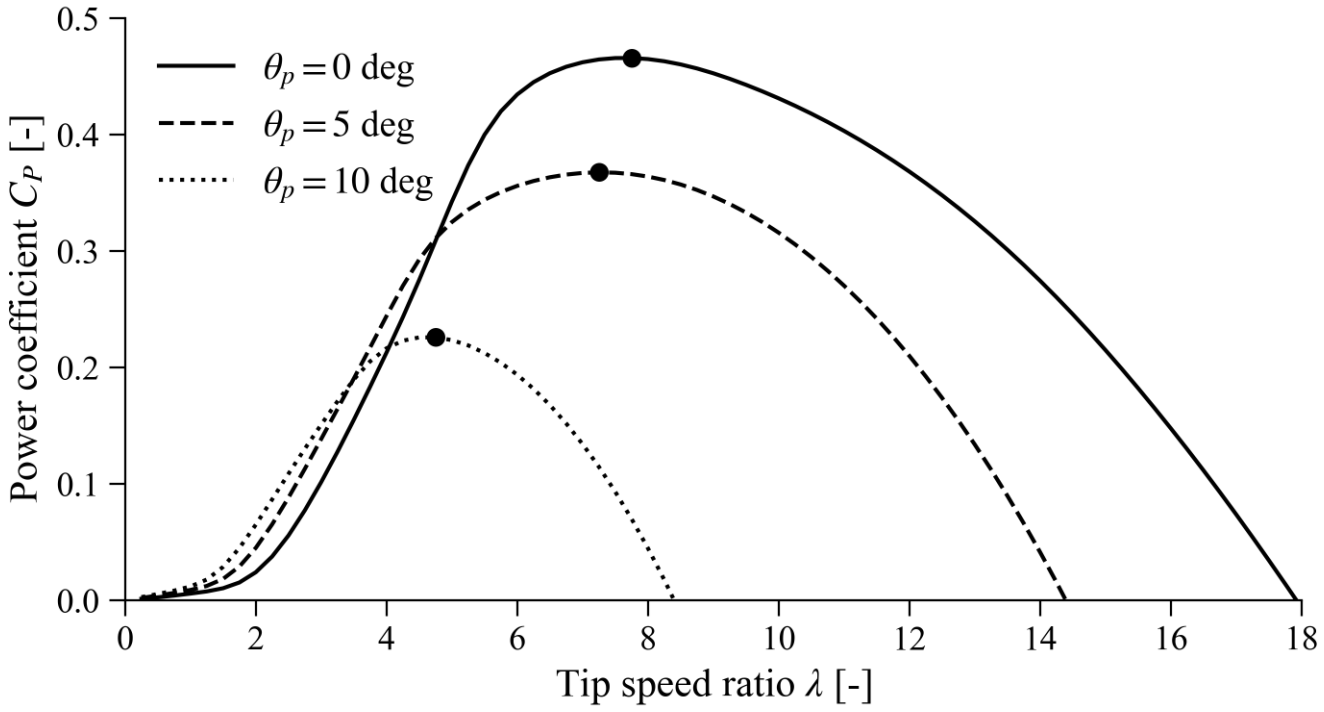
- **ParaView**: for 3D visualization of VTK/VTP files written by OpenFAST
  https://www.paraview.org/
- **pyDatView**: for visualization of input and output files, postprocessing, preliminary scripting
  https://github.com/ebranlard/pyDatView
- **ACDC**: for Campbell diagrams and full structure modes visualization
  https://github.com/OpenFAST/acdc/

Programming:

- **openFAST_toolbox**: (Python) tools for preprocessing and postprocessing OpenFAST files (granular)
  https://github.com/OpenFAST/openfast_toolbox
- **openfast_io** (Python, also ships with openfast), `pip install openfast_io`
- **Matlab-toolbox**: tools for preprocessing and postprocessing OpenFAST files
  https://github.com/OpenFAST/matlab-toolbox
- **WEIS**: tools for preprocessing and postprocessing OpenFAST files (workhorse, e.g. DLC)
  https://github.com/WISDEM/WEIS

Controls

Variation of the power coefficient with TSR and pitch

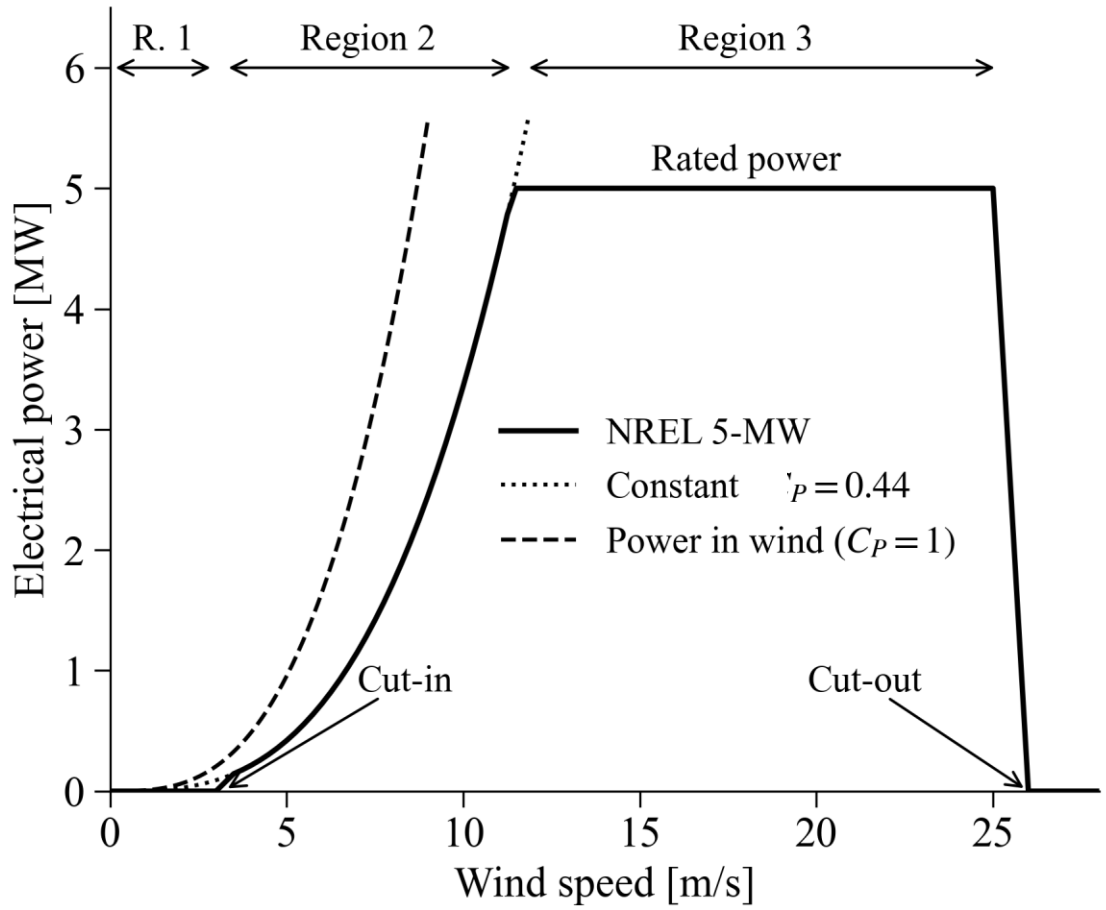


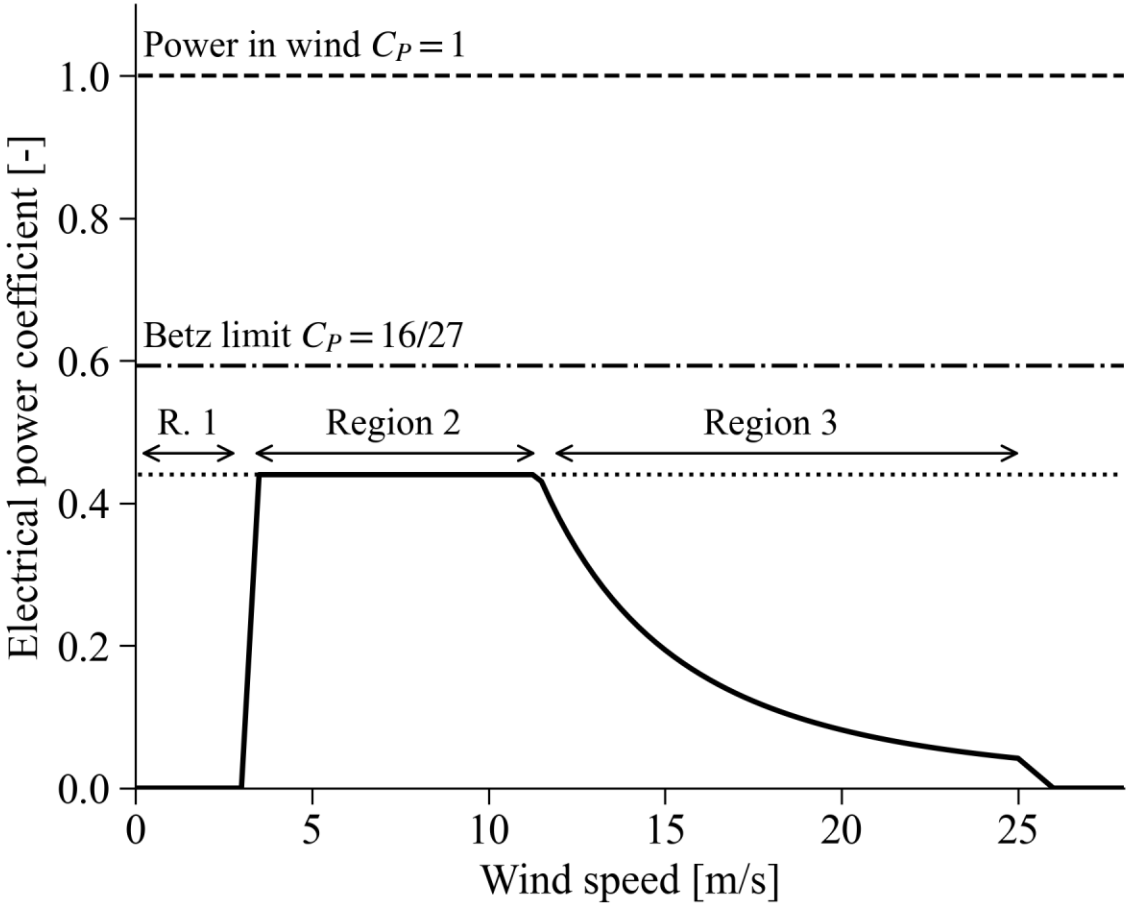
Power curve

The power curve of a given wind turbine is the plot of the power as function of the wind speed. **D**

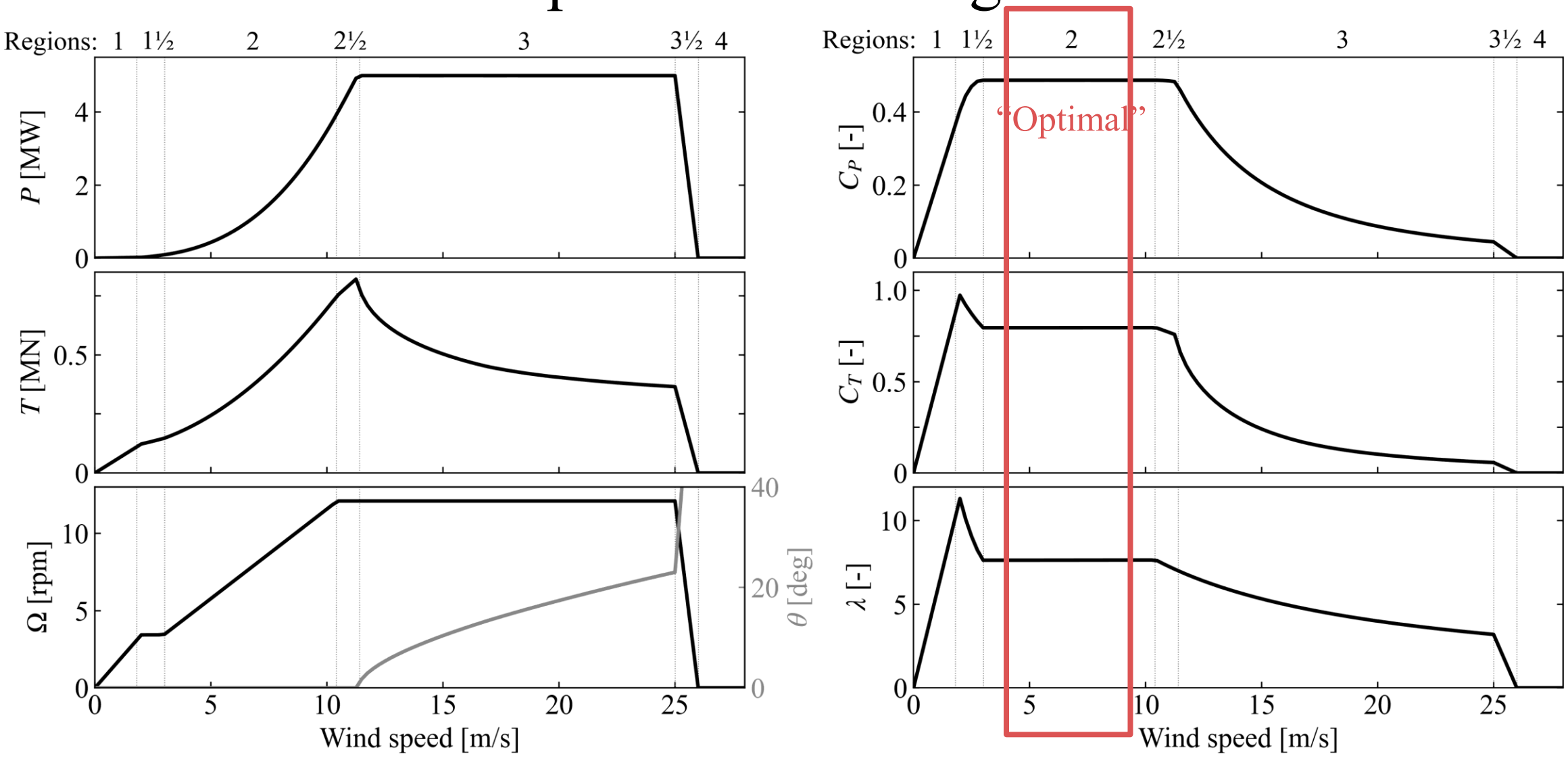
$$\text{Power curve:} \quad P(U_0) \tag{1.6}$$

For a modern wind turbine, the curve follows 3 main regions:





Power curve and other parameters – Region2



In region 2, the turbine typically operates at it's optimal tip-speed ratio, pitch angle and power coefficient:

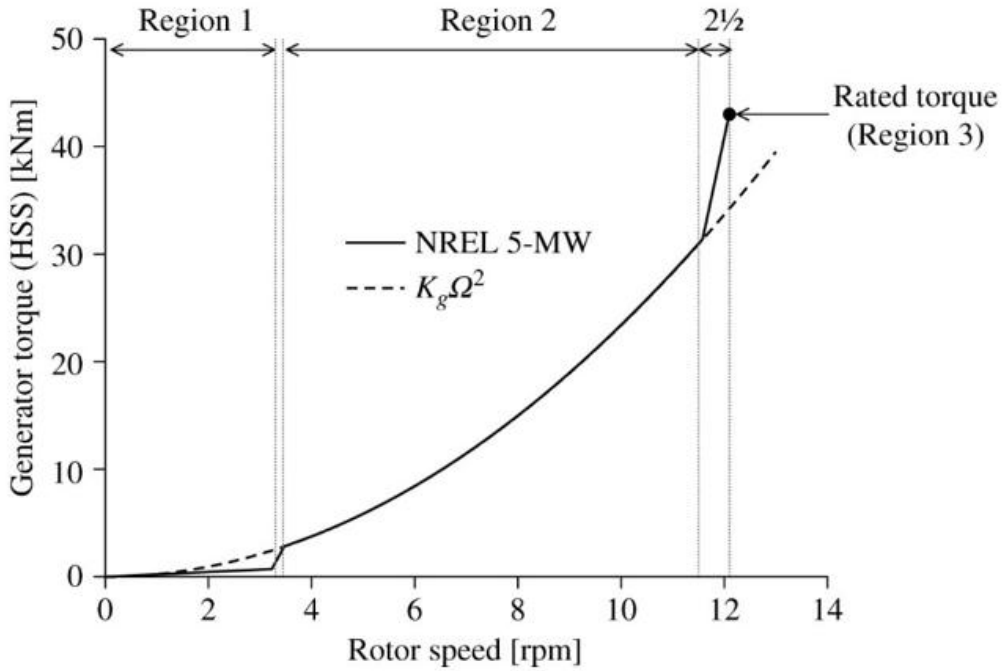
$$\theta_{p,\mathrm{opt}}, \qquad \lambda_{\mathrm{opt}}, \qquad C_{P,\mathrm{opt}}, \tag{1.6}$$

Region 2

Region 2, the generator torque is set to be proportional to the square of the rotational speed of the controller:

$$Q_{g,HSS} = K_g \Omega^2 \tag{9.18}$$

where $Q_{g,HSS}$ is the generator torque on the high-speed shaft (HSS), and K_g is the proportional constant derived in a subsequent paragraph. An example of a generator torque curve as function of the rotor speed is given in Figure 9.14.



Region 2

The optimum pitch, $\theta_{p,opt}$, and the optimal tip-speed ratio, λ_{opt}, are usually chosen as the ones providing the maximum power coefficient. They are therefore defined as:

$$C_{p,\max} = C_p\left(\theta_{p,opt}, \lambda_{opt}\right) \tag{9.19}$$

where C_P is the power coefficient. In Region 2, the generator torque is used to regulate the rotational speed so that λ stays at λ_{opt} over the range of wind speeds of this region.

If the controller successfully achieves the condition $\lambda \equiv \lambda_{opt}$, the aerodynamic power, P_a, is:

$$P_{a,opt} = \frac{1}{2}\rho A U^3 C_P\left(\theta_{opt}, \lambda_{opt}\right) \tag{9.20}$$

The power is seen to be directly proportional to the cube of the wind speed. Similarly, the aerodynamic torque is proportional to the square of the rotational speed:

$$Q_{a,opt} = \frac{P_{a,opt}}{\Omega} = \left[\frac{1}{2}\rho A \frac{R^3}{\lambda_{opt}^3} C_P\left(\theta_{opt}, \lambda_{opt}\right)\right]\Omega^2 = K_a \Omega^2 \tag{9.21}$$

Region 2

$$\text{Ka} = \left[\frac{1}{2}\rho A \frac{R^3}{\lambda_{opt}^3} C_P\left(\theta_{opt}, \lambda_{opt}\right)\right]$$

where K_a is the aerodynamic torque constant with dimension in $\frac{\text{Nm}}{\left(\frac{\text{rad}}{\text{s}}\right)^2}$ If the rotor's speed is given in RPM, the constant is $K_{a,RPM} = K_a\left(\frac{2\pi}{60}\right)^2$.

To achieve a torque balance on the shaft, the generator torque, Q_g, and the aerodynamic torque need to be in equilibrium on the LSS ($Q_g = Q_{a,\,opt}$), therefore:

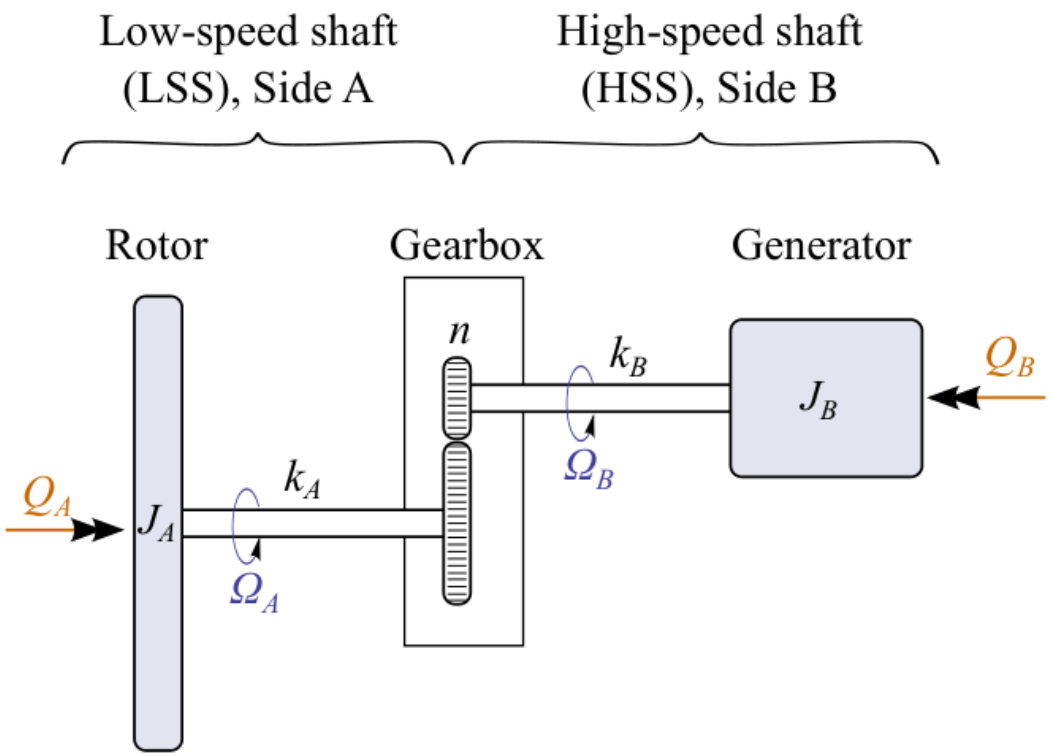
$$Q_g = K_a \Omega^2 \tag{9.22}$$

Using Equation (9.17), the generator torque needed on the HSS is then:

$$Q_{g,HSS} = \frac{\eta}{n_g} Q_g = \frac{\eta}{n_g} K_a \Omega^2 = K_g \Omega^2 \tag{9.23}$$

where K_g is the generator torque constant.

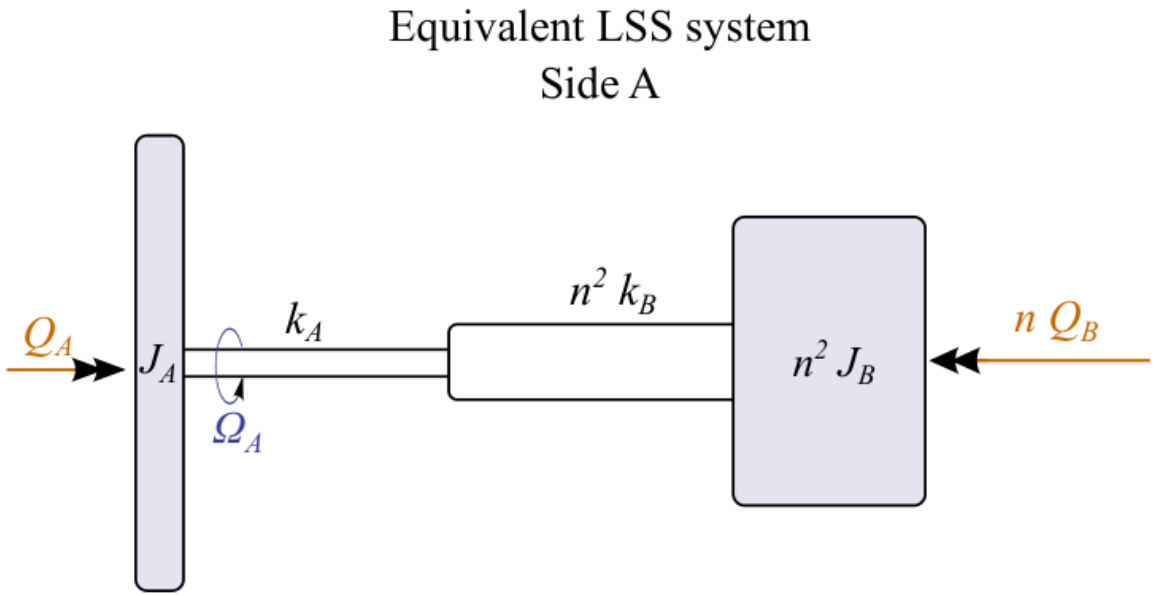
Gears and gear ratio



$$\Omega_B = \Omega_A \cdot n$$
$$Q_A = Q_B \cdot n$$
$$\Omega_A Q_A = \Omega_B Q_B$$

Quantities can be transferred to the other side as follows:

$$J_{B,\text{side A}} = J_B \cdot n^2$$
$$J_{A,\text{side B}} = J_A / n^2$$
$$J_{\text{tot,side A}} = J_A + J_B \cdot n^2$$
$$J_{\text{tot,side B}} = J_B + J_A / n^2$$



Region 2 in OpenFAST

FST:

- CompServo=1

ElastoDyn:

- GenDOF=True

ServoDyn: Simple controller

PCMode=0

- VSContrl = 1 - Simple Variable-speed controller
- GenTiStr - Decide if you want SpdGenOn or TimGenOn
  - At first, use GenTiStr=True and TimGenOn=0
- VS_RtGnSp = **?** - Rated generator speed (HSS side) (rpm)
- VS_RtTq = **?** - Rated generator torque (HSS side) (N-m)
- VS_Rgn2K =**?** - Generator torque constant in Region 2 (HSS side) (N-m/rpm^2)

ServoDyn: more advanced controller, we'll use ROSCO:

- VSContrl = 5

ROSCO

ROSCO (controller): Documentation and examples: https://github.com/NREL/ROSCO

- Compile a controller (using cmake)
- Tune a controller for a given turbine
- See different examples in the repository

Day 3

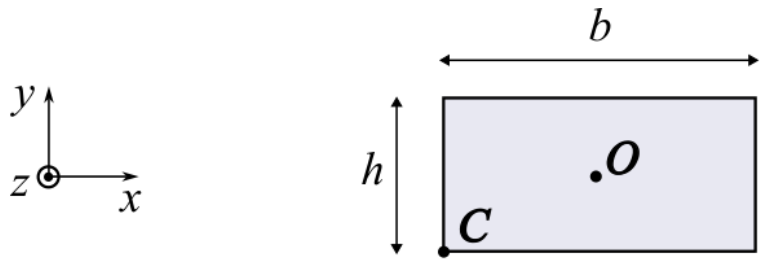
25 MW design – Preliminary design

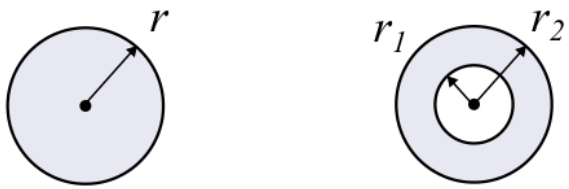
Structure:

- Estimate your maximum thrust force
- Scale the blade stiffness and mass
- Scale the tower stiffness and mass
- Scale some inertias
- See if the model runs…
- Estimate natural frequencies (with your scripts or with linearization, or ACDC). Make sure the tower frequency is not around 1P or 3P
- Estimate the loading needed to have a tower strike.
- Compute the maximum tower top deflection, bending stresses for maximum thrust at the blade root and tower root
- Perform a couple of turbulent simulation, and compute the damage equivalent loads for the blade root bending moment (m=10) and tower bottom bending moment (m=4)

Area moment of inertia – For scaling blade/tower

Analytical integration:





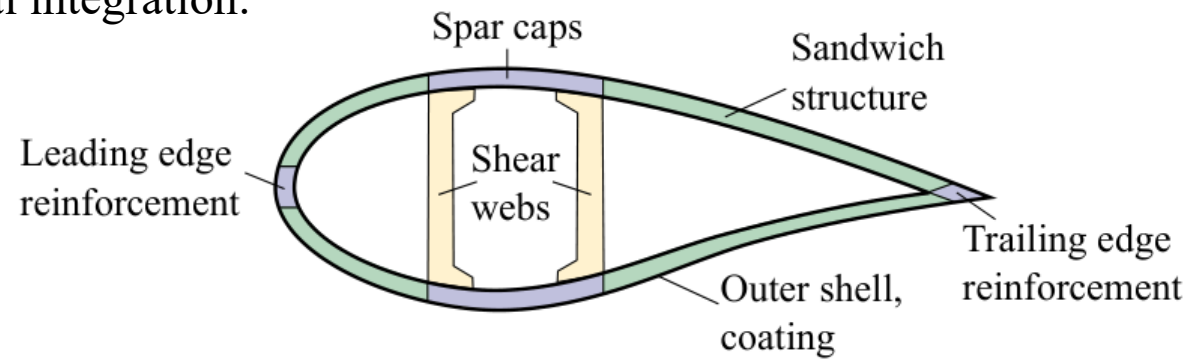
$$I_x = \frac{bh^3}{12}, \qquad I_y = \frac{b^3 h}{12}, \qquad I_p = I_z = \frac{bh}{12}(b^2 + h^2)$$

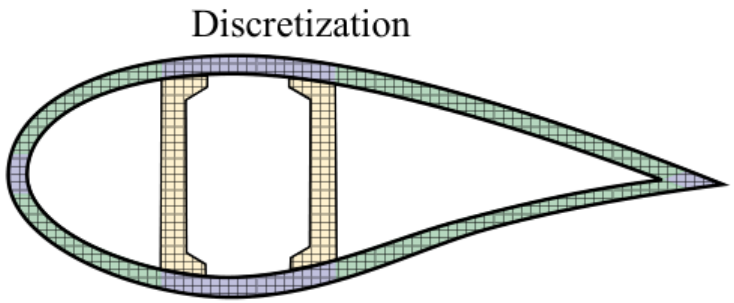
$$I_{x,C} = \frac{bh^3}{3}, \qquad I_{y,C} = \frac{b^3 h}{3}$$

$$I_x = I_y = \frac{1}{4}[r_2^4 - r_1^4], \qquad I_p = I_z = \frac{\pi}{2}[r_2^4 - r_1^4] = 2I_x$$

$$I_x = I_y = \frac{\pi}{64}[D^4 - (D - 2t)^4] \approx \frac{\pi}{8}D^3 t, \qquad I_p = I_z = \frac{\pi}{32}[D^4 - (D - 2t)^4] = 2I_x$$

Numerical integration:





Appendix D - Cuboid

A cuboid is a box-like shape. Assume that a hollow cuboid has width W, length L, and height H, corresponding to x, y, and z axes. The thickness of the walls, top and bottom is t. Then, the volumes, mass, center of gravity, and mass moment of inertia about the y axis are:

$$V = W\,L\,H \qquad V_{in} = W_{in} L_{in} H_{in} \qquad V_{net} = V - V_{in} \tag{D.1}$$

$$m = m_{out} - m_{in} = \rho V_{net} = \rho\left(V - V_{in}\right) \tag{D.2}$$

$$h_{cg} = H / 2 \tag{D.3}$$

$$\begin{aligned} J_{cg} &= \frac{1}{12}\rho\left[V\left(H^2 + W^2\right) - V_{in}\left(H_{in}^2 + W_{in}^2\right)\right] \\ &= \frac{1}{12}\left[m_{out}\left(H^2 + W^2\right) - m_{in}\left(H_{in}^2 + W_{in}^2\right)\right] \end{aligned} \tag{D.4}$$

where $W_{in} = W - 2t$; $L_{in} = L - 2t$; $H_{in} = H - 2t$ and V_{net} is the total material volume.

Appendix D – Hollow cylinder

A hollow cylinder with closed top and bottom with radius R_{out}, height H_{out}, and wall thickness t has the following volume, mass, center of gravity, and mass moment of inertia:

$$V = \pi R^2 H \qquad V_{in} = \pi R_{in}^2 H_{in} \tag{D.5}$$

$$V_{net} = V - V_{in} \tag{D.6}$$

$$m = m_{out} - m_{in} = \rho V_{net} = \rho\pi\left(R^2 H - R_{in}^2 H_{in}\right) \tag{D.7}$$

$$h_{cg} = H / 2 \tag{D.8}$$

$$J_{cg} = \frac{1}{12}\rho\pi\left[R^2 H\left(3R^2 + H^2\right) - R_{in}^2 H_{in}\left(3R_{in}^2 + H_{in}^2\right)\right] \tag{D.9}$$

where $R_{in} = R - t$ and $H_{in} = H - 2t$. If the cylinder is open, then $H_{in} = H_{out}$.

Appendix D- Frustrum

A frustum is essentially the lower part of a cone. Consider a hollow frustum with open top and bottom which has an outer upper radius R_A, outer lower radius R_B, upper wall thickness t_A , lower wall thickness t_B, and height H.

The volumes are defined by the outer and inner radii:

$$V = \frac{\pi H}{3}\left(R_A^2 + R_A R_B + R_B^2\right) \qquad V_{in} = \frac{\pi H}{3}\left(R_{A,in}^2 + R_{A,in} R_{B,in} + R_{B,in}^2\right) \tag{D.10}$$

$$V_{net} = V - V_{in} \tag{D.11}$$

The mass is then:

$$m = m_{out} - m_{in} = \rho V_{net} = \rho\left(V - V_{in}\right) \tag{D.12}$$

The centers of gravity $h_{cg,out}$ of the outer frustum and inner frustum $h_{cg,in}$ are given from Equation (D.13):

$$h_{cg,out} = \frac{H\left(3R_A^2 + 2R_A R_B + R_B^2\right)}{4\left(R_A^2 + R_A R_B + R_B^2\right)} \qquad h_{cg,in} = \frac{H\left(3R_{A,in}^2 + 2R_{A,in} R_{B,in} + R_{B,in}^2\right)}{4\left(R_{A,in}^2 + R_{A,in} R_{B,in} + R_{B,in}^2\right)} \tag{D.13}$$

The center of gravity of the frustum is found from:

$$h_{cg} = \frac{h_{cg,out} V - h_{cg,in} V_{in}}{V} \tag{D.14}$$

8.8.7 Maximum Stresses

Maximum bending stresses, σ_b, in the blade root due to flapwise moments applied to the blade, M, are related to the thickness of the root, t, and its area moment of inertia, I, by $\sigma_b = M(t/2)/I = M/(2I/t)$, as should be clear from discussions in Chapter 4. (The root is assumed to be solid here).

To determine how bending stresses change with rotor size, scaling laws for the area moment of inertia need to be determined. For simplicity, consider the blade root to be approximated by a rectangular cross-section of width c (corresponding to the chord) and thickness t. The area moment of inertia about the flapping axis is $I = ct^3/12$. If the radius is doubled, then the moment of inertia goes up by a factor of 16, and the thickness by a factor of 2. The ratio $2I/t$, which is given by $2I/t = ct^2/6$, is then increased by a factor of 8, just like the aerodynamic moments. In general, the blade root area moment of inertia scales as R^3.

Scaling laws

Table 8.11 Summary of scaling relations

Quantity	Symbol	Relation	Scale dependence
Power, forces, and moments			
Power	P	$P_1/P_2 = (R_1/R_2)^2$	$\sim R^2$
Torque	Q	$Q_1/Q_2 = (R_1/R_2)^3$	$\sim R^3$
Thrust	T	$T_1/T_2 = (R_1/R_2)^2$	$\sim R^2$
Rotational speed	Ω	$\Omega_1/\Omega_2 = (R_1/R_2)^1$	$\sim R^{-1}$
Weight	W	$W_1/W_2 = (R_1/R_2)^3$	$\sim R^3$
Aerodynamic moments	M_A	$M_{A,1}/M_{A,2} = (R_1/R_2)^3$	$\sim R^3$
Centrifugal forces	F_c	$F_{c,1}/F_{c,2} = (R_1/R_2)^2$	$\sim R^2$
Stresses			
Gravitational	σ_g	$\sigma_{g,1}/\sigma_{g,2} = (R_1/R_2)^1$	$\sim R^1$
Aerodynamic	σ_A	$\sigma_{A,1}/\sigma_{A,2} = (R_1/R_2)^0 = 1$	$\sim R^0$
Centrifugal	σ_c	$\sigma_{c,1}/\sigma_{c,2} = (R_1/R_2)^0 = 1$	$\sim R^0$
Resonances			
Natural frequency	ω	$\omega_{n,1}/\omega_{n,2} = (R_1/R_2)^1$	$\sim R^{-1}$
Excitation	Ω/ω	$(\Omega_1/\omega_{n,1})/(\Omega_2/\omega_{n,2}) = (R_1/R_2)^0 = 1$	$\sim R^0$

Note: R, rotor radius

Wind Turbine Ideal Scaling Relations

Important observations:

- "Square-cube law"
  - **Power** increases with radius **squared**
  - **Weight** increases with Radius **cubed**
  - (Real world not that bad so far)
- Most **stresses** do **<u>not</u>** change
- Blade **stress** due to **gravity** **<u>increases</u>** with radius
- Natural **frequencies** **<u>decrease</u>** with radius

Scaling - Old

NREL National Renewable Energy Laboratory

A national laboratory of the U.S. Department of Energy
Office of Energy Efficiency & Renewable Energy

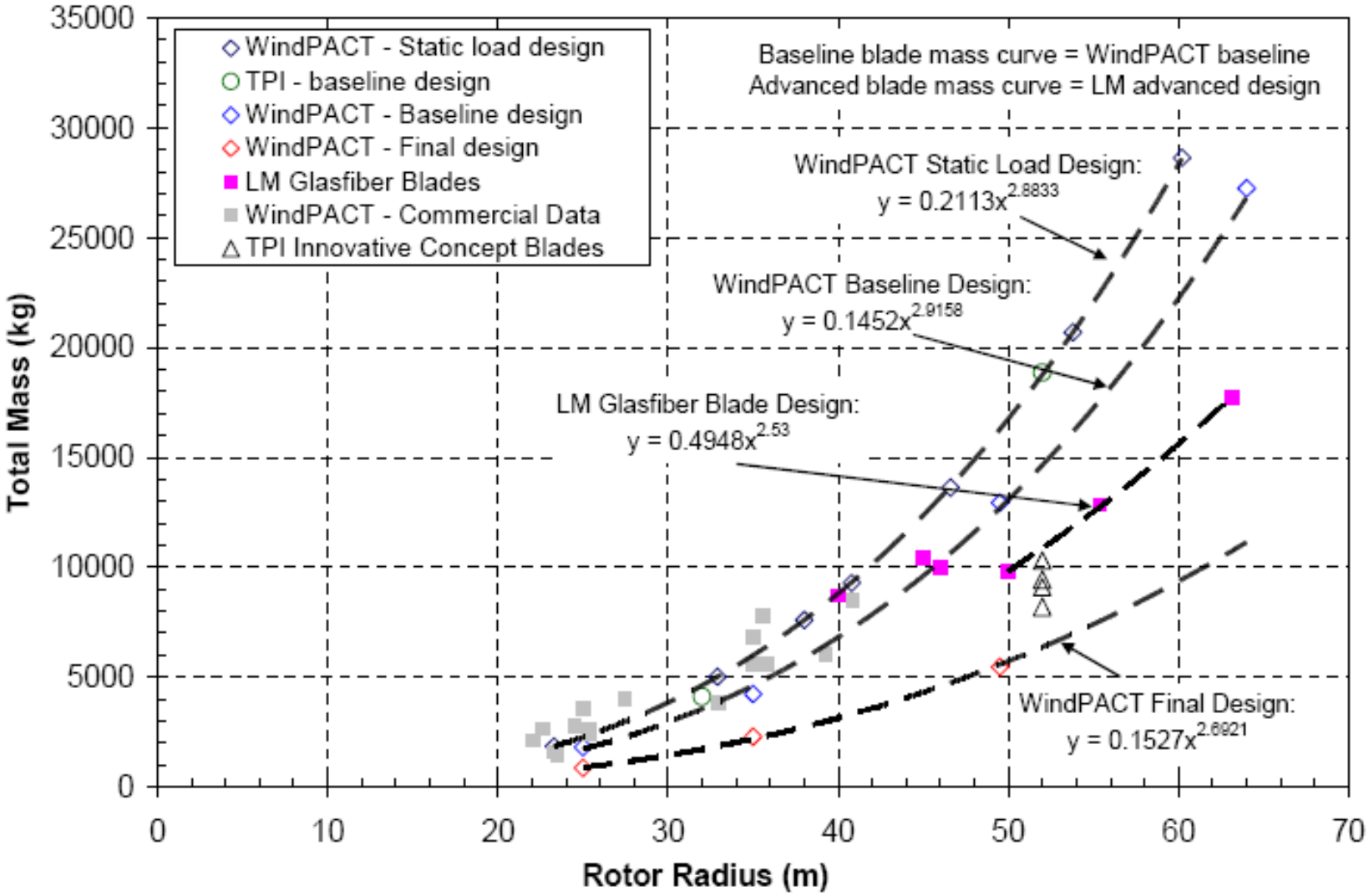
Innovation for Our Energy Future

Wind Turbine Design Cost and Scaling Model

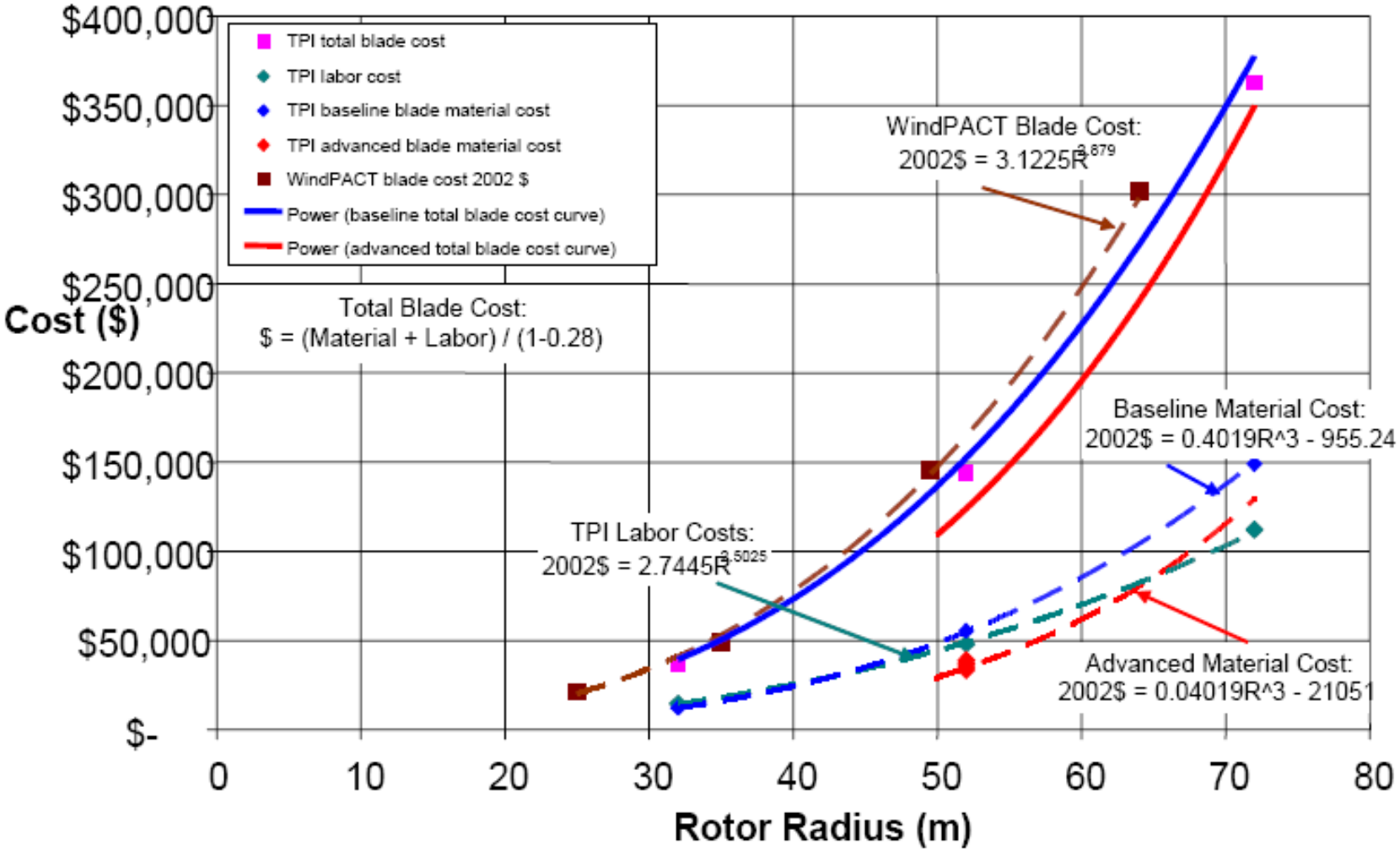
L. Fingersh, M. Hand, and A. Laxson

Technical Report
NREL/TP-500-40566
December 2006

Blade Mass Scaling



Blade Cost Scaling



Blade mass and stiffness scaling

IEA-22-280-RWT.yaml

```yaml
14  blade:
15    outer_shape_bem:
16      airfoil_position:
17        grid: [0.0, 0.02, 0.04868313468735217, 0.06649308419703373, 0.083455918
18        labels: [circular, circular, FB90, FB80, FB70, FB60, SNL-FFA-W3-560FB, S
19      chord:
20        grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07, 0.
21        values: [5.800000000000002, 5.80000750475248, 5.8000282508983085, 5.8733
22      twist:
23        grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07, 0.
24        values: [0.20943951023931956, 0.2094346686429543, 0.2094224788066553, 0.
25      pitch_axis:
26        grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07, 0.
27        values: [0.5, 0.4999812305251822, 0.4996464935337481, 0.4955917730476608
28      reference_axis:
29        x:
30          grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07,
31          values: [0.0, -3.0625000000000003e-07, -1.12e-06, -7.250467567039361e-
32        y:
33          grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07,
34          values: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0
35        z:
36          grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07,
37          values: [-8.077935669512783e-28, 1.378, 2.756, 4.1339999999999995, 5.5
38      rthick:
39        grid: [0.0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.060000000000000005, 0.07, 0.
40        values: [1.0, 0.9999970648471761, 0.999988951343826, 0.9822867352930104,
```

Blade mass and stiffness scaling

Alternative: use these simple rules:

Potential equations for estimating blade structural parameters, given the chord and the thickness. They are based on the NREL 5 MW, reverse engineered by Sandia, and described here: Definition of a 5MW/61.5m Wind Turbine Blade Reference Model, SAND2013-2569, 2013, where c = chord and t = thickness.

For flapwise stiffness
$(EI)_f = 3.5 \times 10^6 ct^3$ [Nm^2]

For edgewise stiffness:
$(EI)_e = 6.5 \times 10^6 c^3 t$ [Nm^2]

For mass density:
$m = 120ct$ [kg/m]

For torsional stiffness
$(EI)_\tau = 2(EI)_e$ [Nm^2]

For edgewise inertia (more or less matches)
$I_e = 4 \times 10^{-7}(EI)_e$ [m^4]

For flapwise inertia
$I_f = 4.0 \times 10^{-7}(EI)_f$ [m^4]

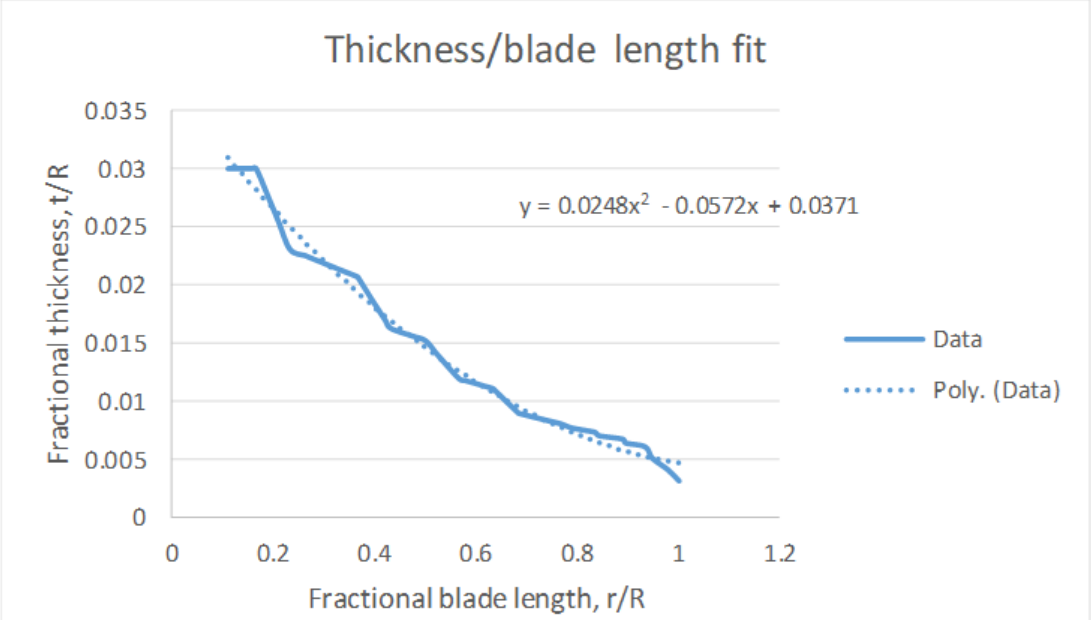
For axial stiffness (this is very approximate)
$(EI)_{axial} = 1.28 \times 10^7\, m$ [N]

The chord is determined primarily from aerodynamic considerations

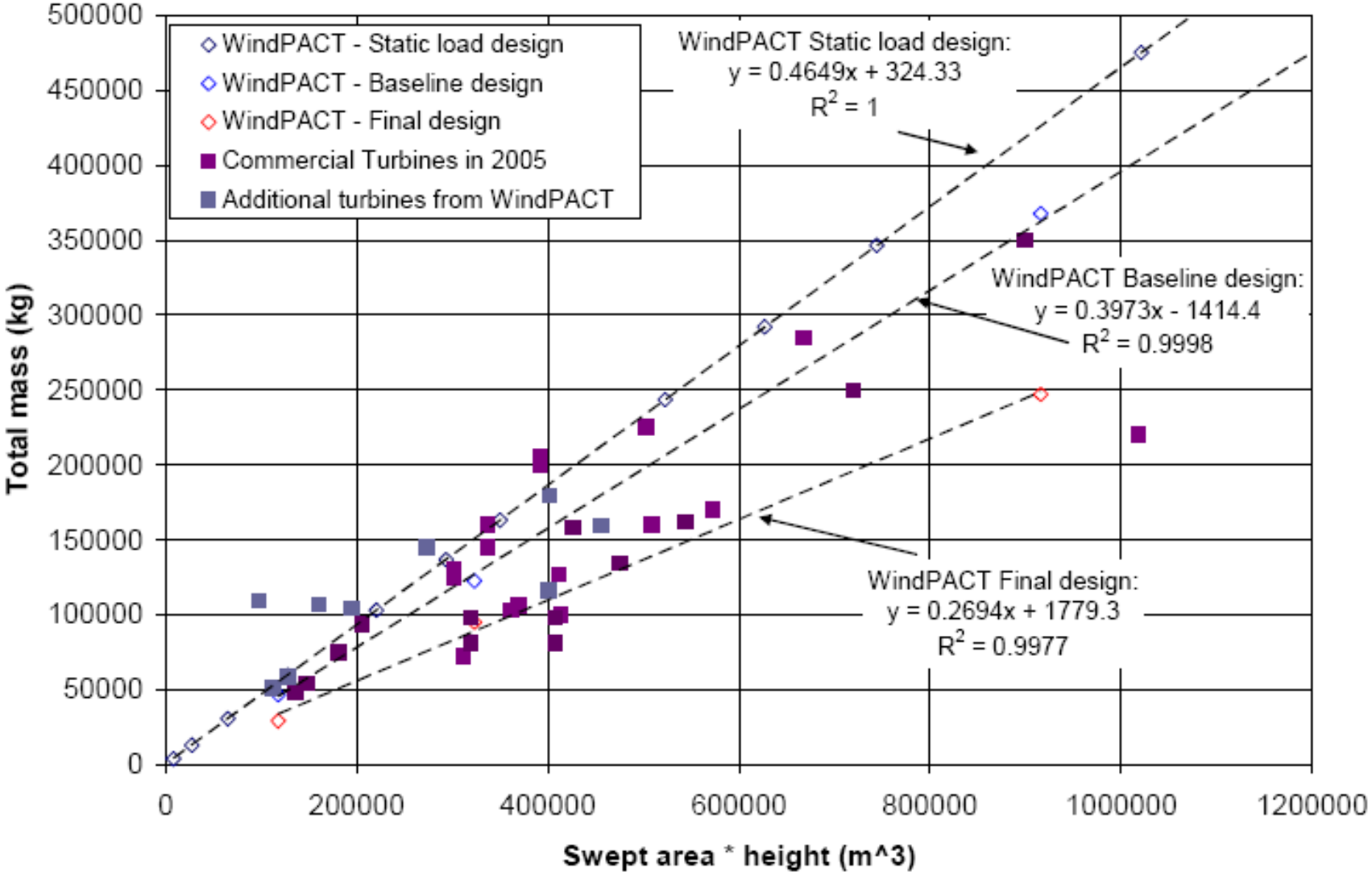
Also the thickness as function of radius is given approximately by:

$$t = R[0.0248x^2 - 0.572x + 0.0371]$$

Where $x = r/R$



Tower Mass Scaling



Tower diameter and thickness

Use: chord, thickness, and I of a cuboic/hollow cuboid -

IEA-22-280-RWT.yaml

```yaml
tower:
  outer_shape_bem:
    reference_axis: &id003
      x:
        grid: [0.0, 0.021339950396958213, 0.05508684863374078, 0.08883374687052335, (
        values: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.(
      y:
        grid: [0.0, 0.021339950396958213, 0.05508684863374078, 0.08883374687052335, (
        values: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.(
      z:
        grid: [0.0, 0.021339950396958213, 0.05508684863374078, 0.08883374687052335, (
        values: [15.0, 18.18788983, 23.22920397, 28.27051811, 33.31183226, 38.353146
    outer_diameter:
      grid: [0.0, 0.021339950396958213, 0.05508684863374078, 0.08883374687052335, 0.
      values: [10.0, 10.0, 10.0, 9.912094021695989, 9.798867733731491, 9.68321254942
    drag_coefficient:
      grid: [0.0, 1.0]
      values: [0.5, 0.5]
  internal_structure_2d_fem:
    reference_axis: *id003
    layers:
      - name: tower_wall
        material: steel
        thickness:
          grid: [0.0, 0.021339950396958213, 0.05508684863374078, 0.08883374687052335
          values: [0.06693667948991197, 0.0657205223939668, 0.06351493329530639, 0.0(
```

Example: 1.5 MW Land Based Turbine

From Input Page	
Machine Rating (kWs)	1500
Rotor Diameter (meters)	70
Hub Height (meters)	65

Land Based Turbine		
Cost in $	2002	
Component	Component Costs $1000	Component Mass kgs
Rotor	**237**	**28,291**
Blades	152	13,845
Hub	43	10,083
Pitch mchnsm & bearings	38	3,588
Spinner, Nose Cone	4	775
Drive train,nacelle	**617**	**43,556**
Low speed shaft	21	3,025
Bearings	12	679
Gearbox	153	10,241
Mech brake, HS cpling etc	3	
Generator	98	5,501
Variable spd electronics	119	
Yaw drive & bearing	20	1,875
Main frame	93	19,763
Electrical connections	60	
Hydraulic, Cooling system	18	120
Nacelle cover	21	2,351
Control, Safety System, Condition Monitoring	**35**	
Tower	**147**	**97,958**
	0	
TURBINE CAPITAL COST (TCC)	**1,036**	**169,804**

Example (2)

Foundations	46	
Transportation	50	
Roads, Civil Work	79	
Assembly & Installation	38	
Electrical Interface/Connections	122	
Engineering & Permits	32	
	0	
	0	
	0	
BALANCE OF STATION COST (BOS)	367	0
	0	
Initial capital cost (ICC)	1,403	169,804
Installed Cost per kW (cost in $)	935	113,203
Turbine Capital per kW sans BOS & Warranty (cost in $)	691	113,203
Levelized Replacement Cost $ per year	16	
O&M $ per turbine/yr	30	
Land Lease Cost	5	
CAPACITY FACTOR	32.82%	
Net ANNUAL ENERGY PRODUCTION Energy MWh (AEP)	4312	
Fixed Charge Rate	11.85%	
COE $/kWh	0.0476	

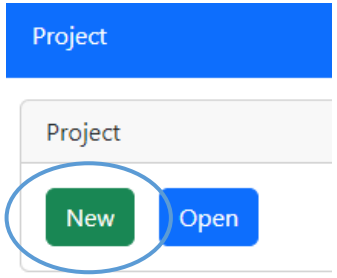
Linearization – Case 12_Campbell_ACDC

ACDC: Automated Campbell Diagram Code

https://github.com/OpenFAST/acdc

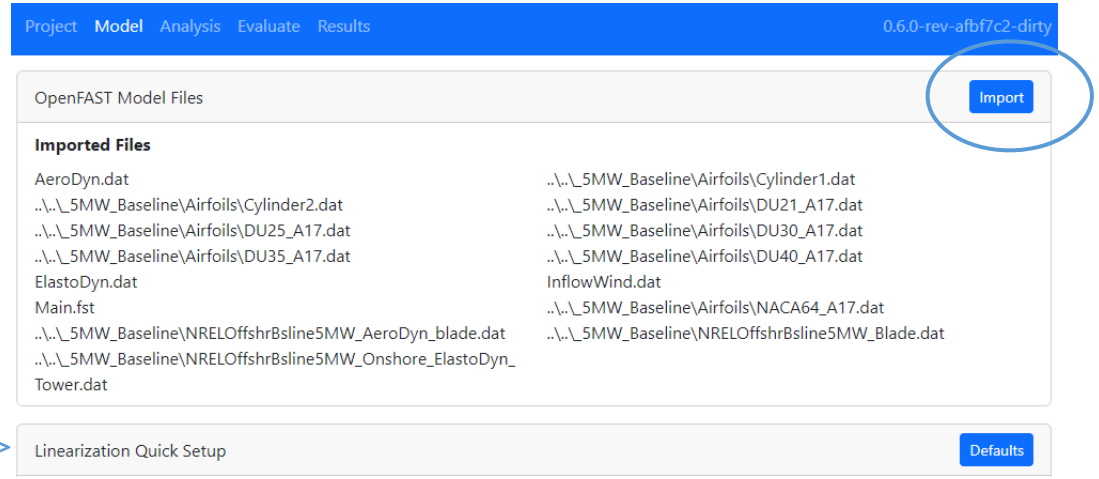
https://github.com/OpenFAST/acdc/releases

Project tab: Create a new project (will save a "`.json`" file)



Model Tab:

- Import `Main.fst` from your template folder `Simulations\Template_NoServo`
- Make sure: `OutFmt` is large enough (e.g. `ES20.11E2`)
- If you updates the files, ACDC won't know it.
- I recommend editing files and then reimporting instead of changing them from the GUI



Look but don't edit ->

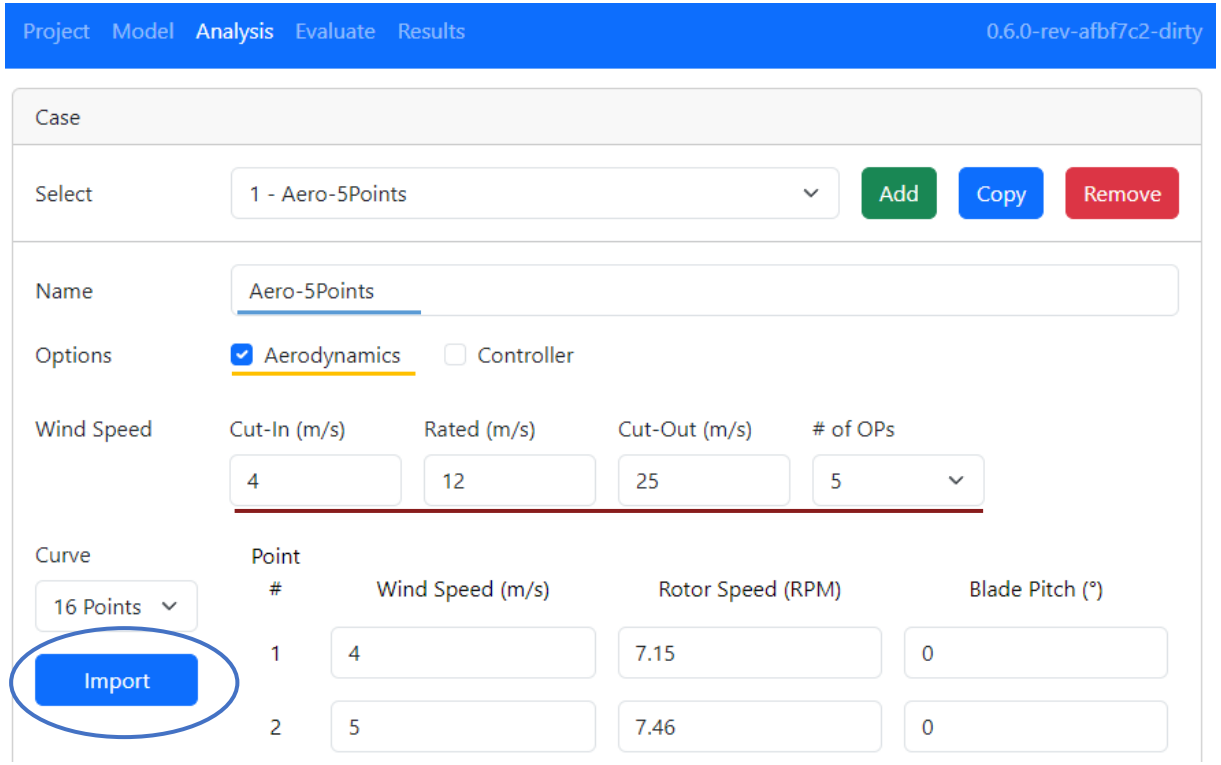
Linearization – Case 12_Campbell_ACDC

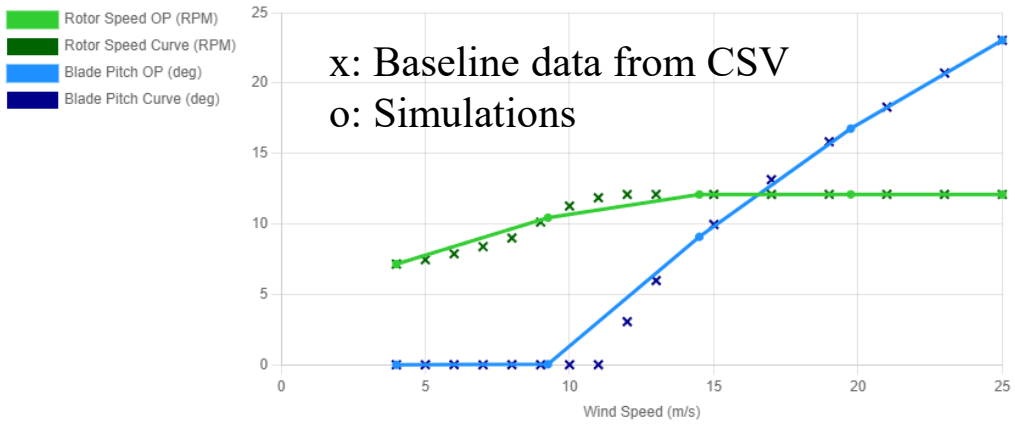
Analysis tab:

- Choose a Name (optional)
- Choose if Aerodynamics or controller is included.
- Set Wind Speed range and number Operating points (OPs) (number of simulations run)
- Manually insert "Baseline points" OR Import CSV file "`OperatingPoints.csv`"

OperatingPoints.csv

```
#WindSpeed_[m/s] , RotSpeed_[rpm] , Pitch_[deg]
4.00             , 7.15           , 0.00
5.00             , 7.46           , 0.00
6.00             , 7.88           , 0.00
7.00             , 8.39           , 0.00
8.00             , 9.00           , 0.00
9.00             , 10.13          , 0.00
10.00            , 11.27          , 0.00
11.00            , 11.86          , 0.00
12.00            , 12.10          , 3.06
13.00            , 12.10          , 5.98
15.00            , 12.10          , 9.96
17.00            , 12.10          , 13.15
19.00            , 12.10          , 15.84
21.00            , 12.10          , 18.30
23.00            , 12.10          , 20.72
25.00            , 12.10          , 23.05
```

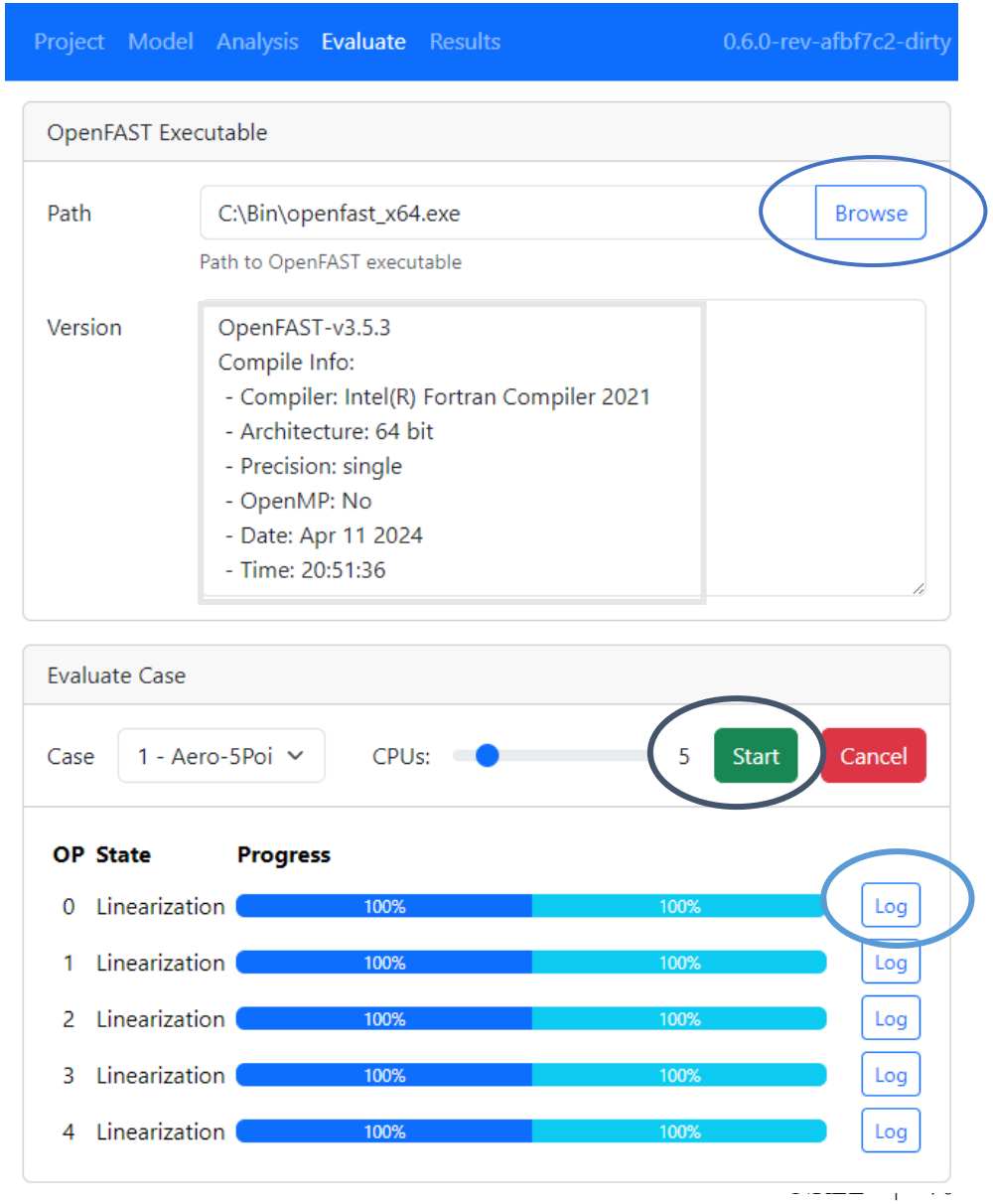




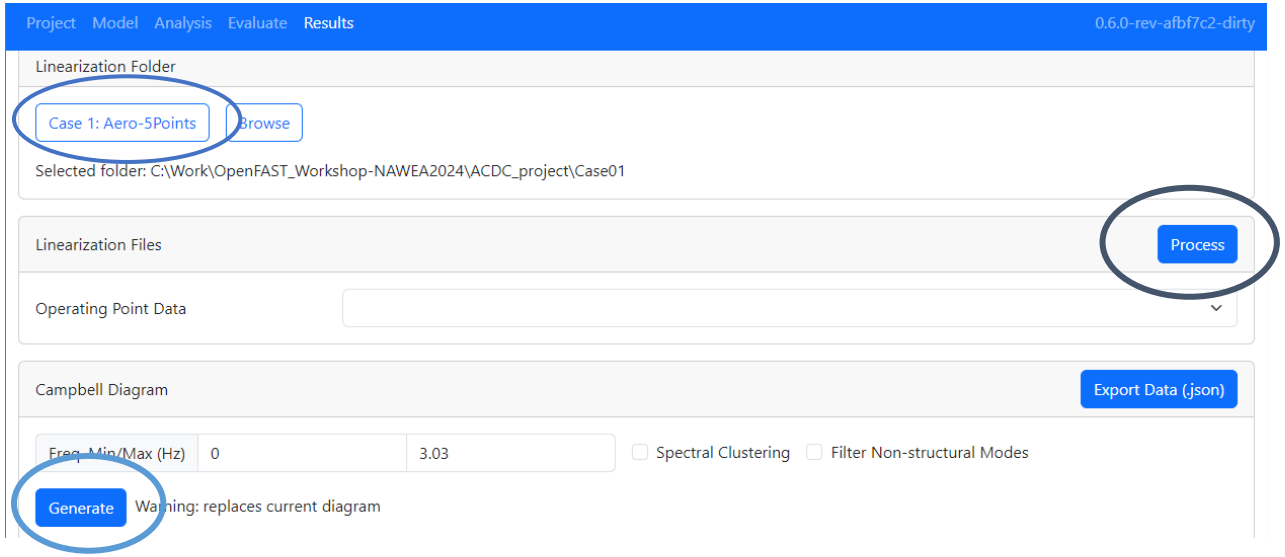
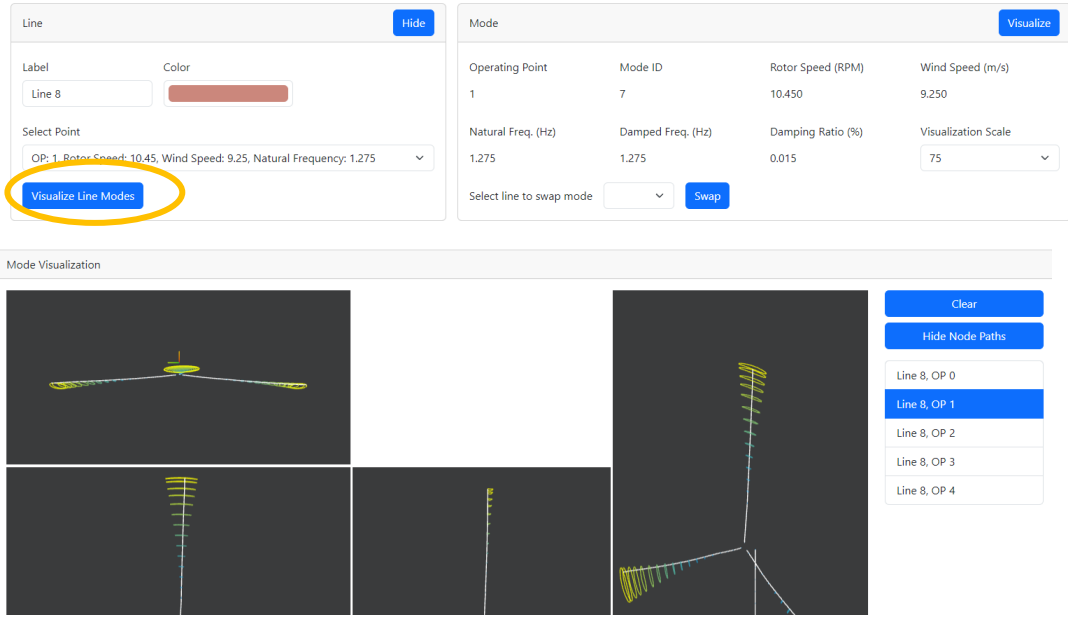
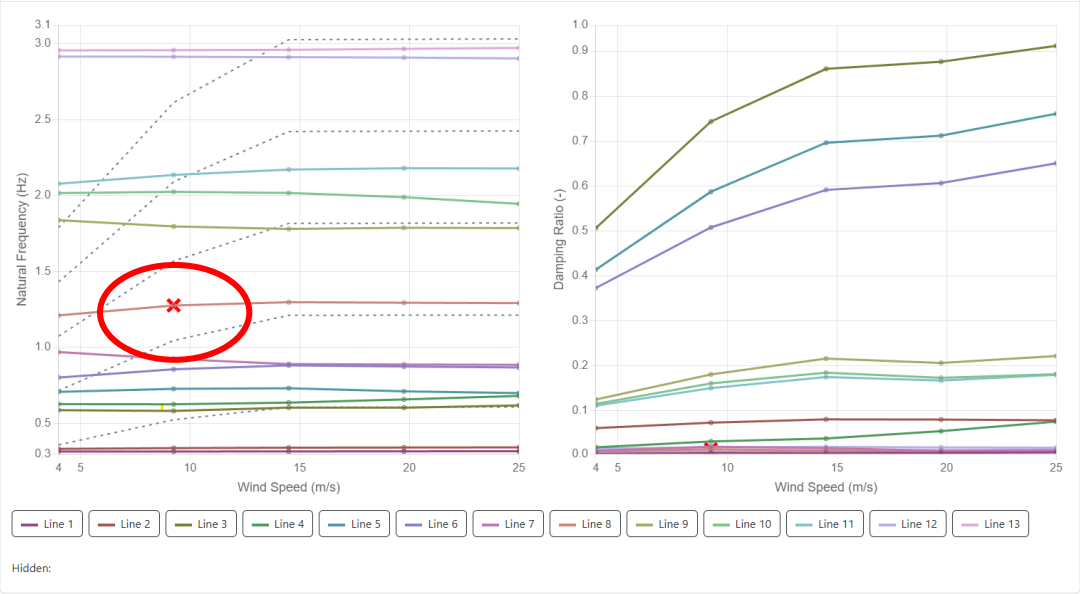
Linearization – Case 12_Campbell_ACDC

Evaluate tab:

- Choose an OpenFAST executable (compatible with ACDC, v3.5)
- Make sure "Version info" is displayed. Otherwise. Make sure the openfast executable run on your system.
- Adapt number of CPUs (processors) to your machine capabilities and start the simulations.
- Check the logs if errors occur.



Linearization – Case 12_Campbell_ACDC

Results tab:

- Choose a case
- Process results
- Generate Campbell diagram
- Select a Point on Campbell Diagram
- Click on Visualize Line Modes